

Design and Deployment of a VMware vSphere Virtual Infrastructure

Track system administration

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Project Requirements

- Install at least two ESXi hosts as virtual machines within workstation
- Deploy vCenter Server as a virtual appliance (with version you choose)
- Connect your ESXi hosts to vCenter Server.
- Combine your ESXi host into a cluster to enable advanced features like vMotion, High Availability (HA), and Distributed Resource Scheduler (DRS)
- Create a basic virtual machine and install an operating system (Windows or Linux)
- Upload ISOs to a content library
- Perform operations like cloning, snapshotting, and template creation on your VM
- Migrate VMs between ESXi hosts
- Create and configure virtual switches (vSwitches) where one is for the management and VM Port Groups and another switch which is for storage and migration
- Create and manage NFS datastore
- Configure HA and DRS, try to make one esxi fails, to make sure if the VM will start on the other esxi.
- Enable FT on the VM

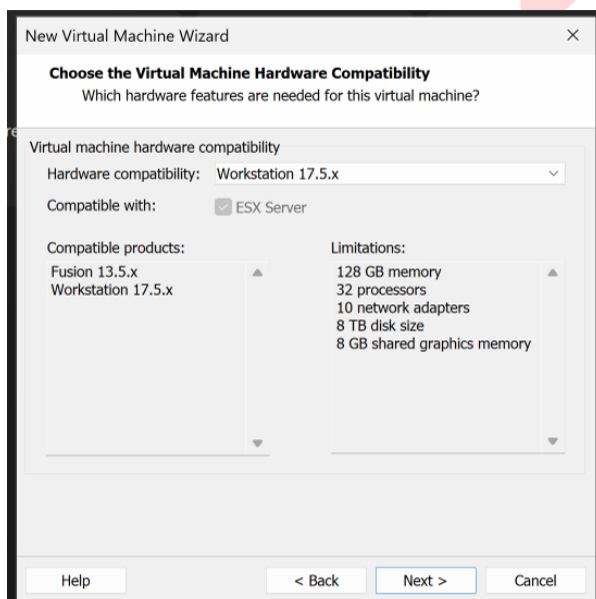
1. Install at least two ESXi hosts as virtual machines within workstation

The following is a demonstration of an ESXi installation on one of our local machines. The ESXi was deployed using 2 processors, 4 cores per processor, and 12 GB RAM.

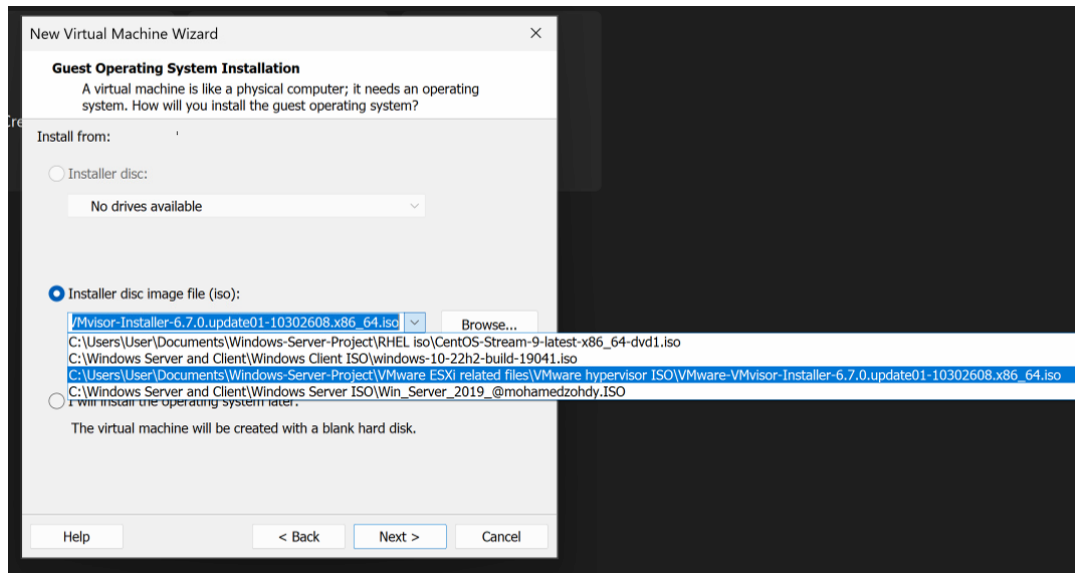
1- Select Custom (Advanced)



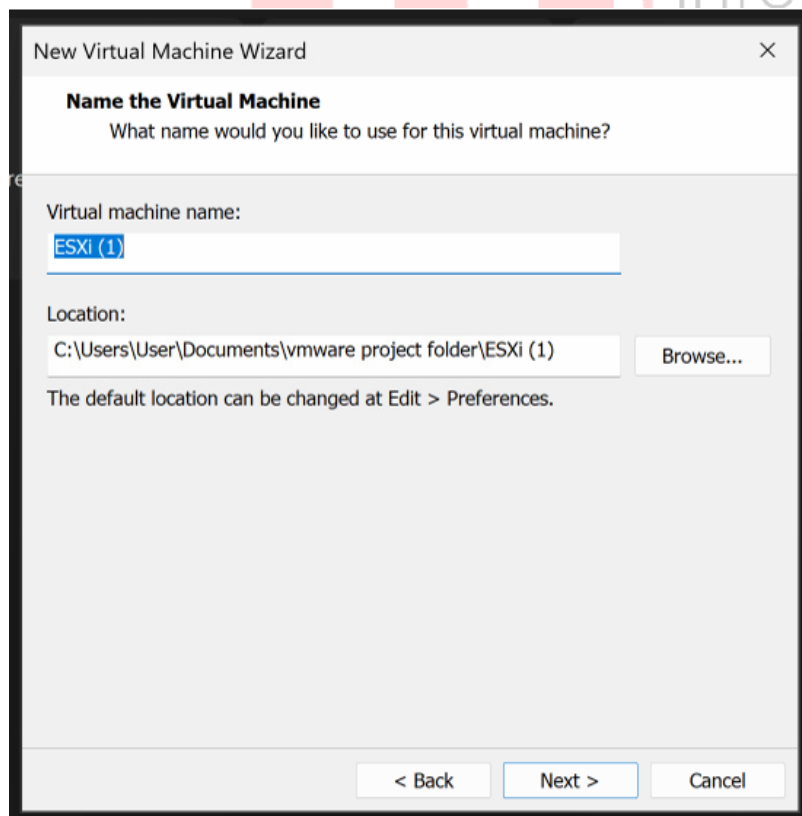
2- Choosing the virtual machine hardware compatibility



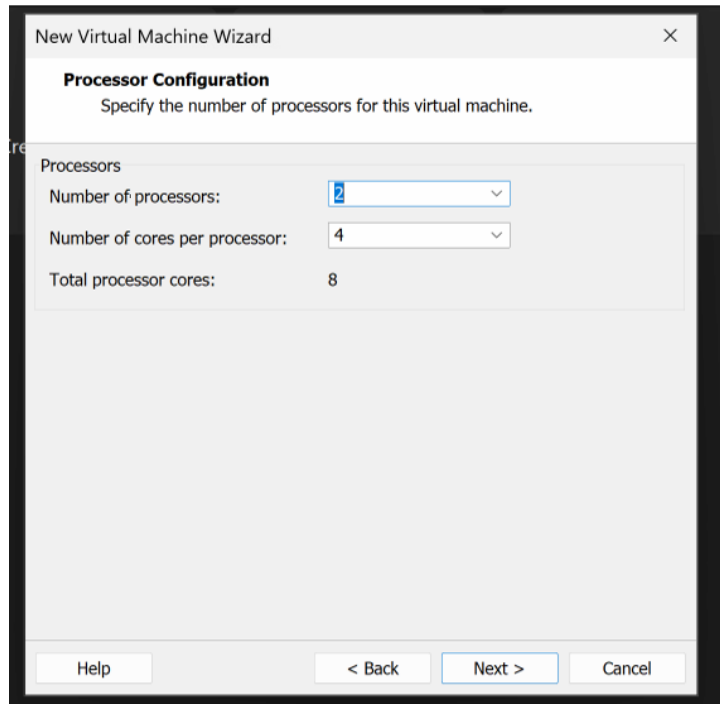
3- Selecting the VMware hypervisor ISO



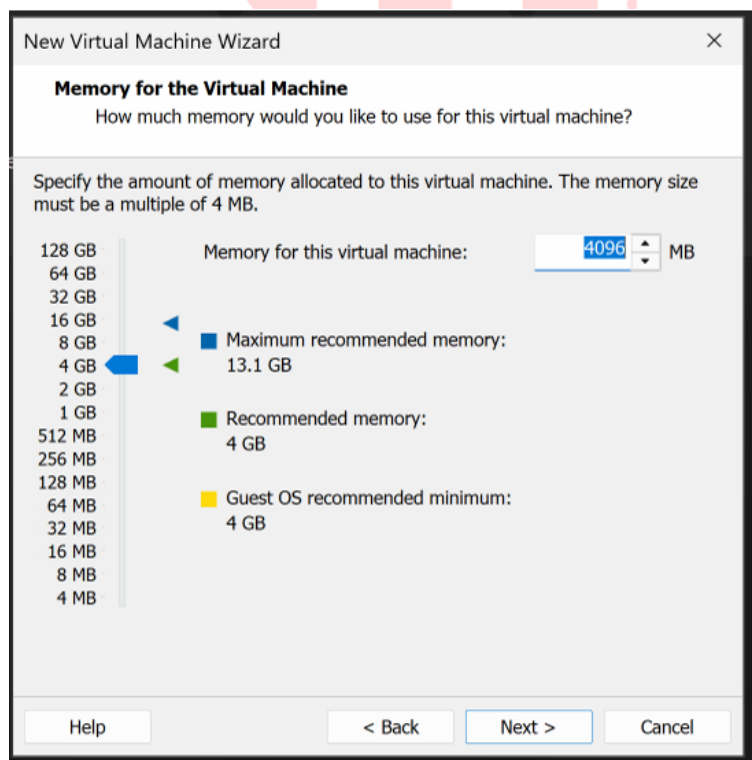
4- Renaming the ESXi host and location of the VM files



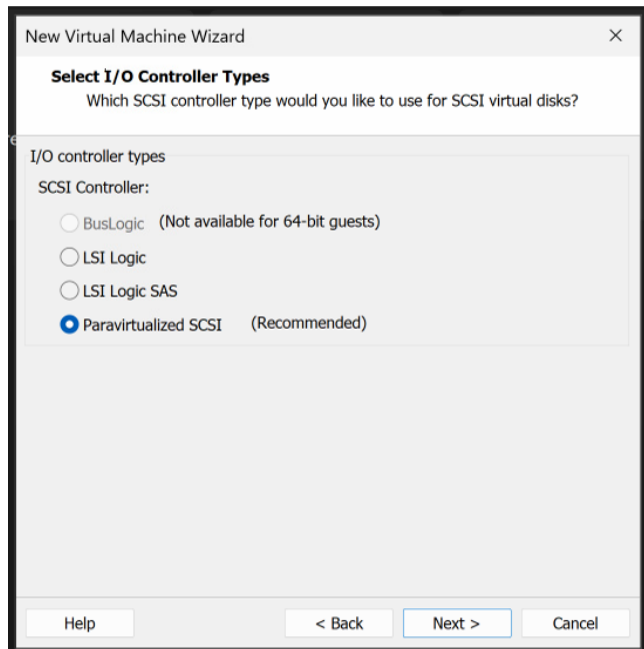
5- Configuring the processor



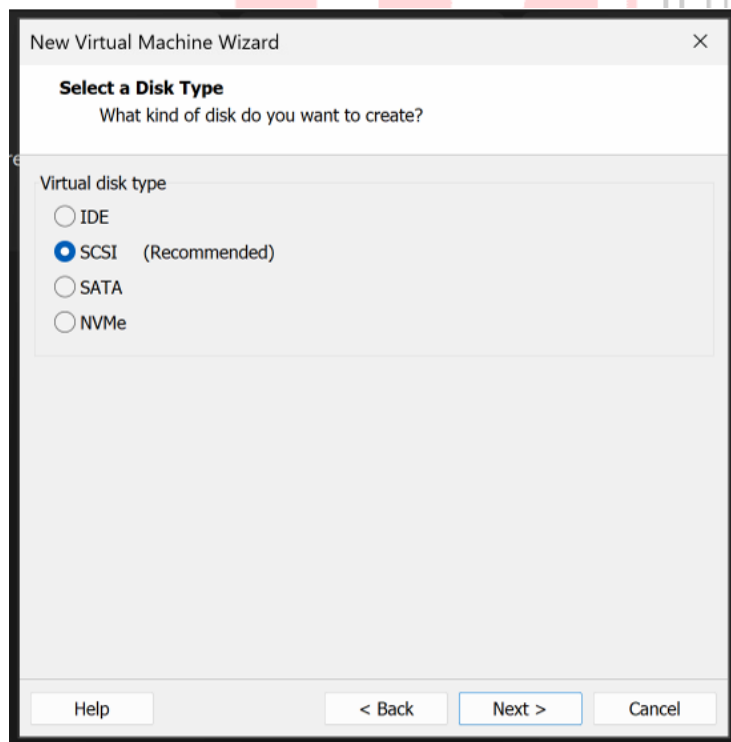
6- Set the RAM to 12288 MB or 12 GB RAM. (The value displayed has Changed)



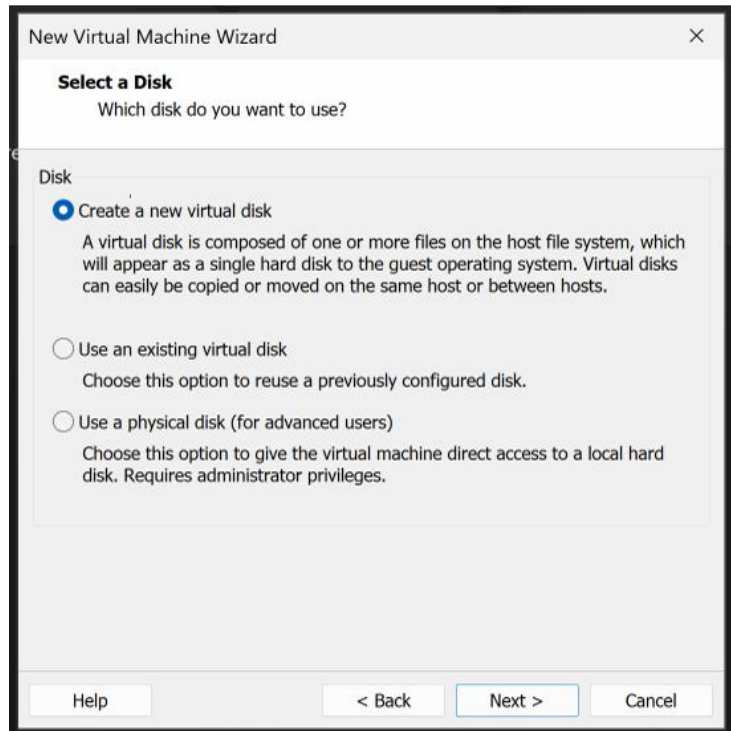
7- We will be using the recommended paravirtualized SCSI as I/O controller type since it provides better performance and lower CPU overhead.



8- Selected the Virtual disk type as SCSI



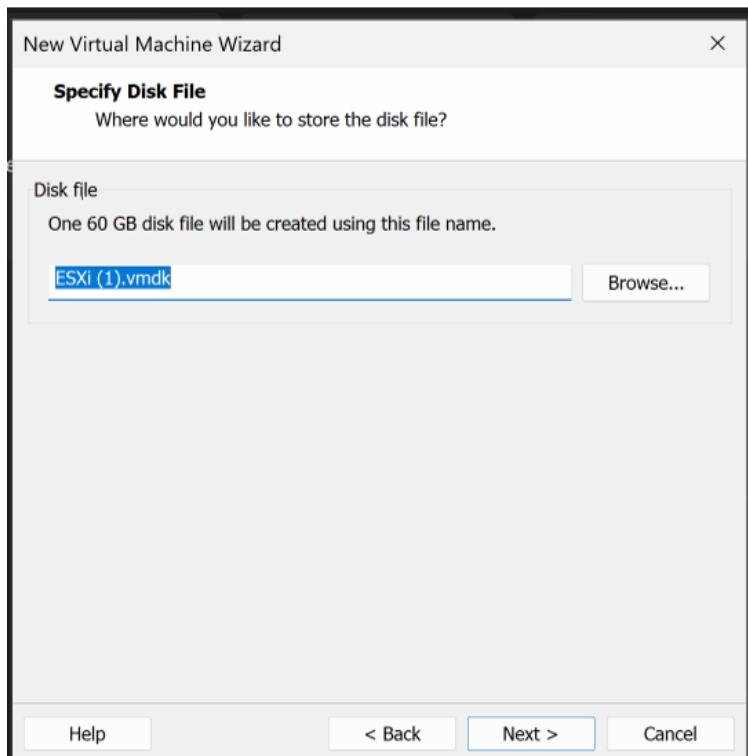
9- Selected a new virtual disk



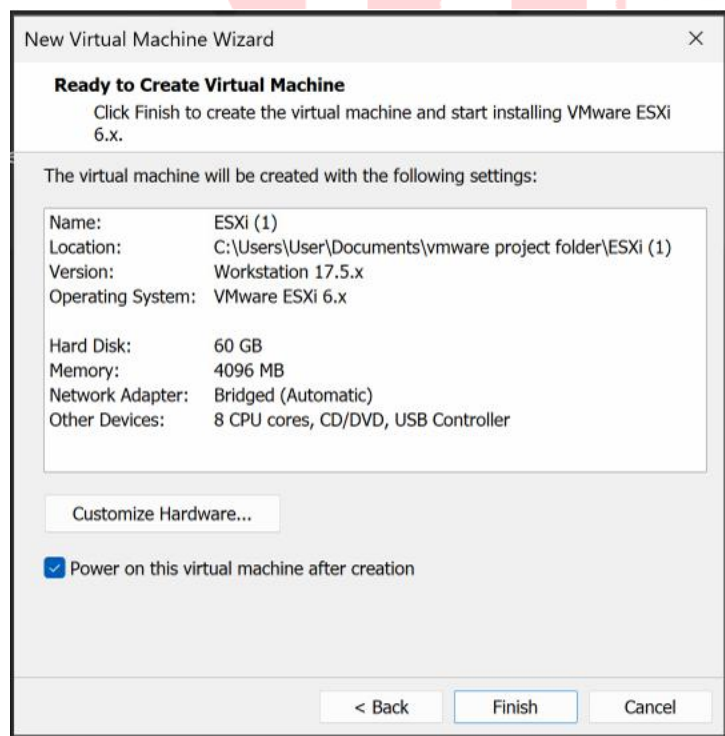
10- Specified the Disk capacity



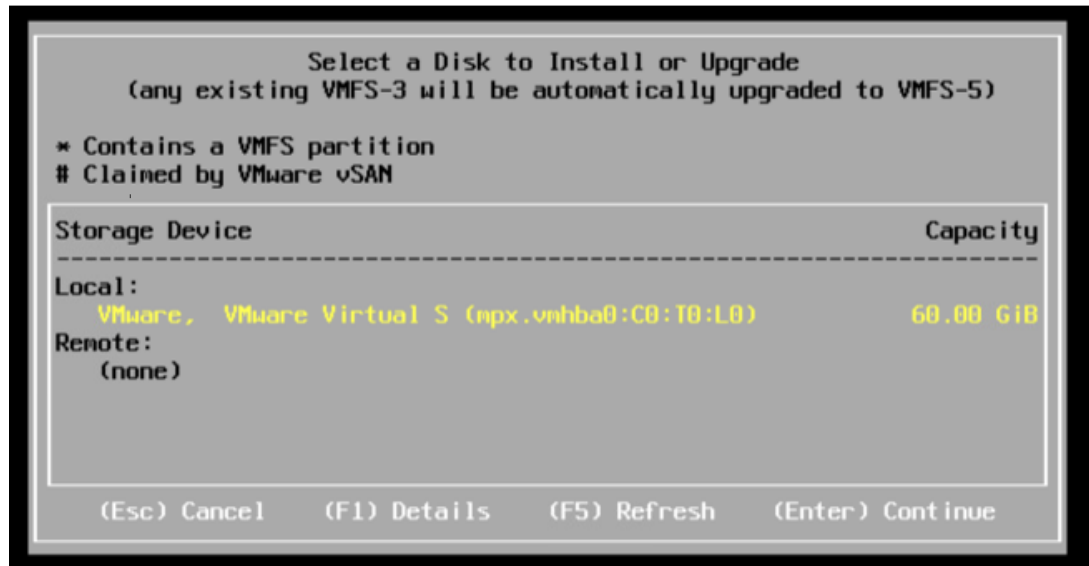
11- Renaming the Disk File as ESXi (1).vmdk



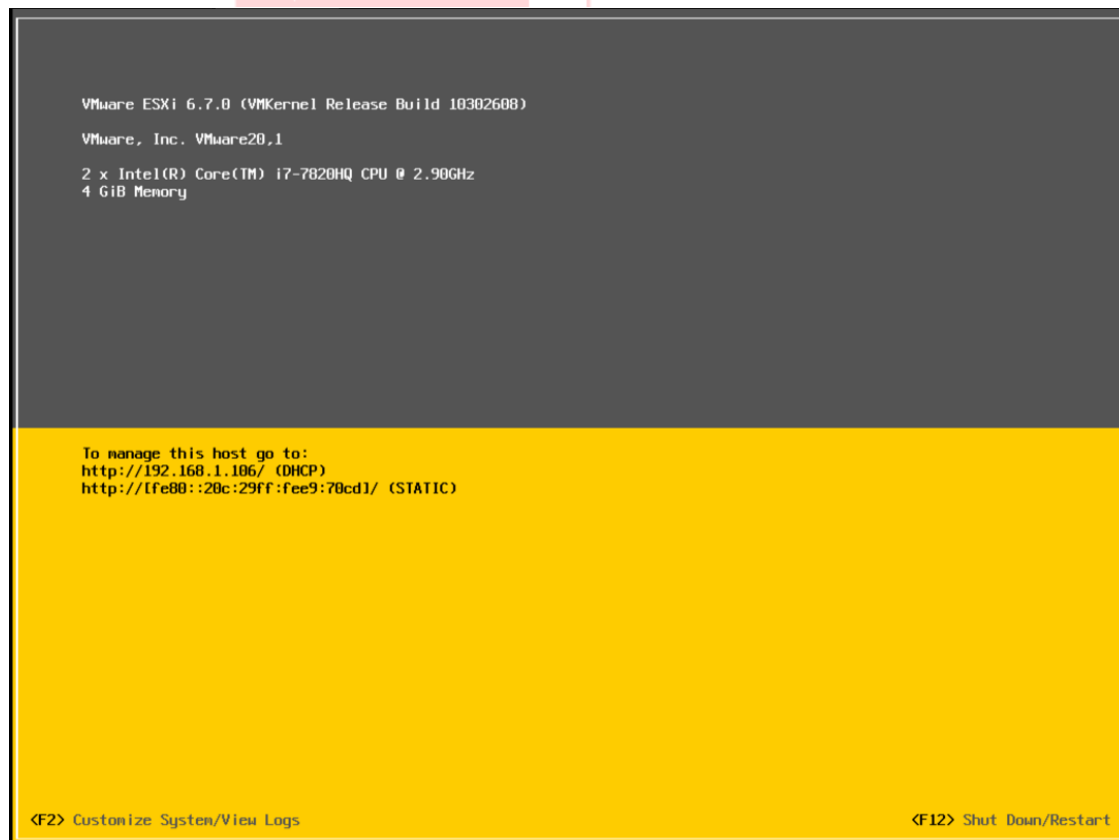
12- Confirming the final steps



13- Selecting the storage device to install the ESXi 6.7



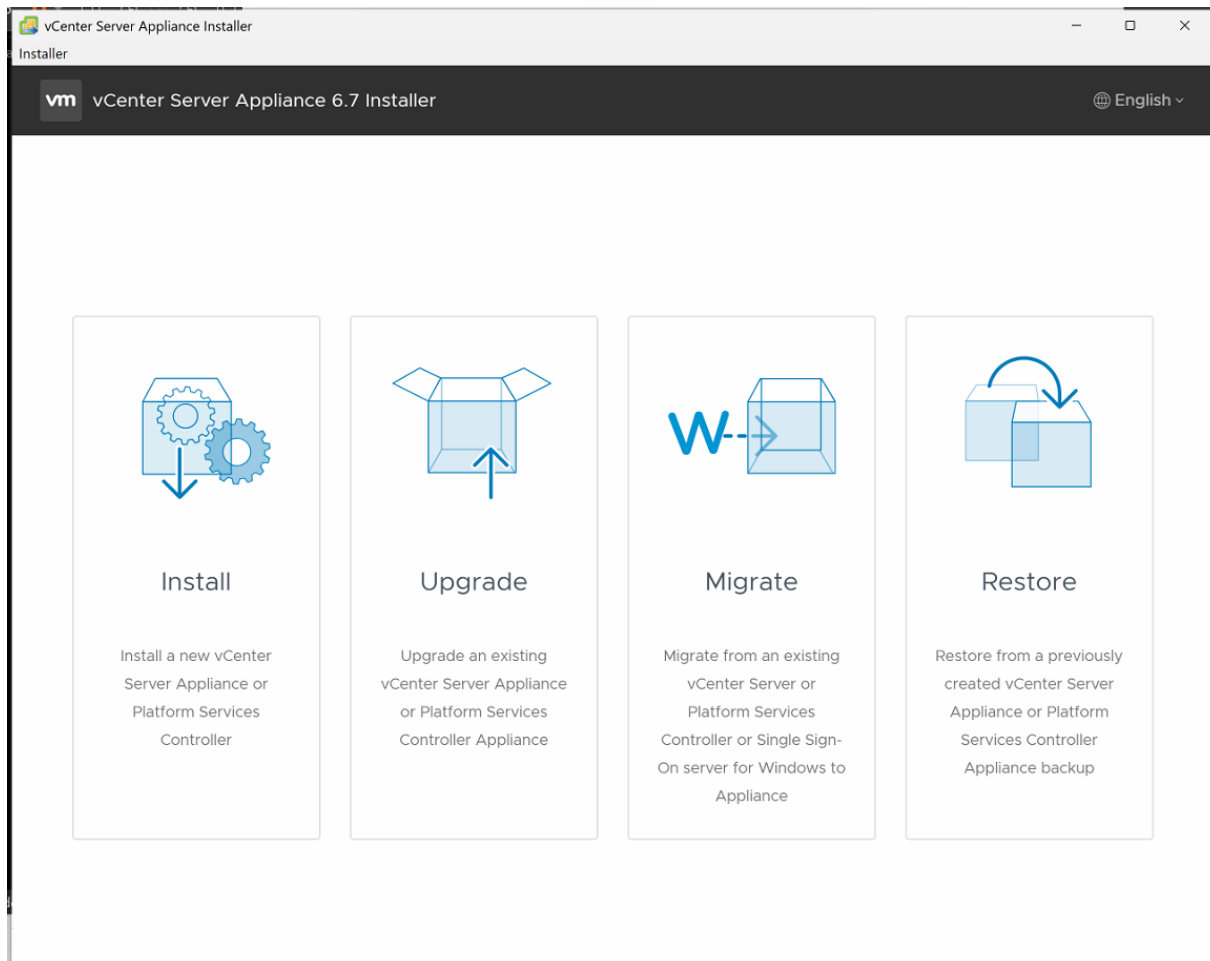
14- Opening the ESXi installation



2. Deploy vCenter Server as a Virtual appliance (with version you choose)

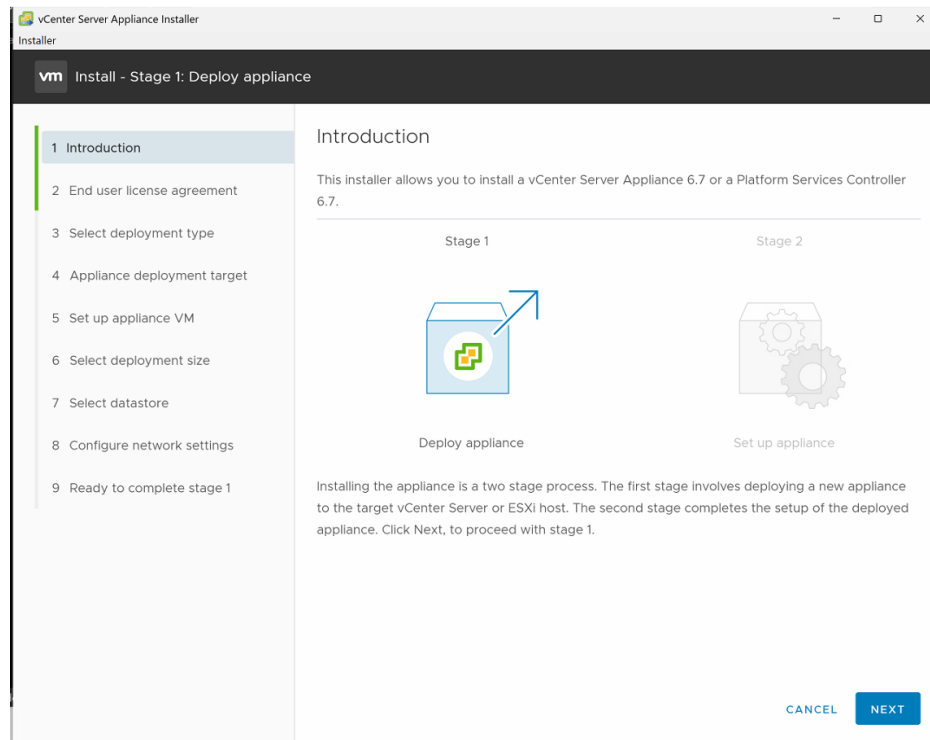
We have downloaded the VCSA 6.7 ISO and did the following

- Select mount from the ISO
- Opened the installer folder / win32
- Clicked on the installer.exe
- Opened the vCenter Server Appliance 6.7 Installer

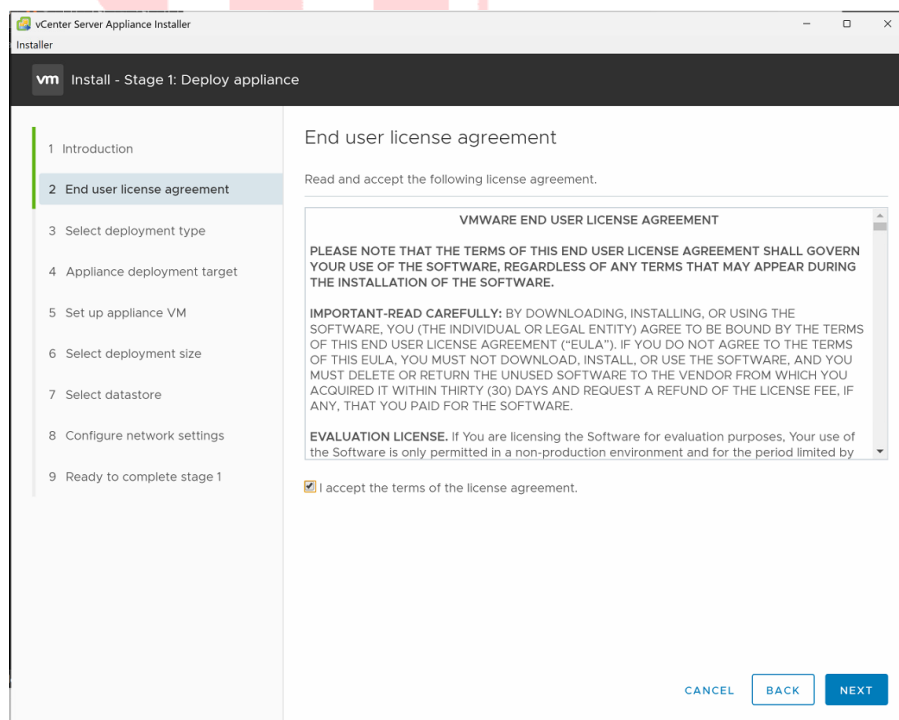


Stage-1:

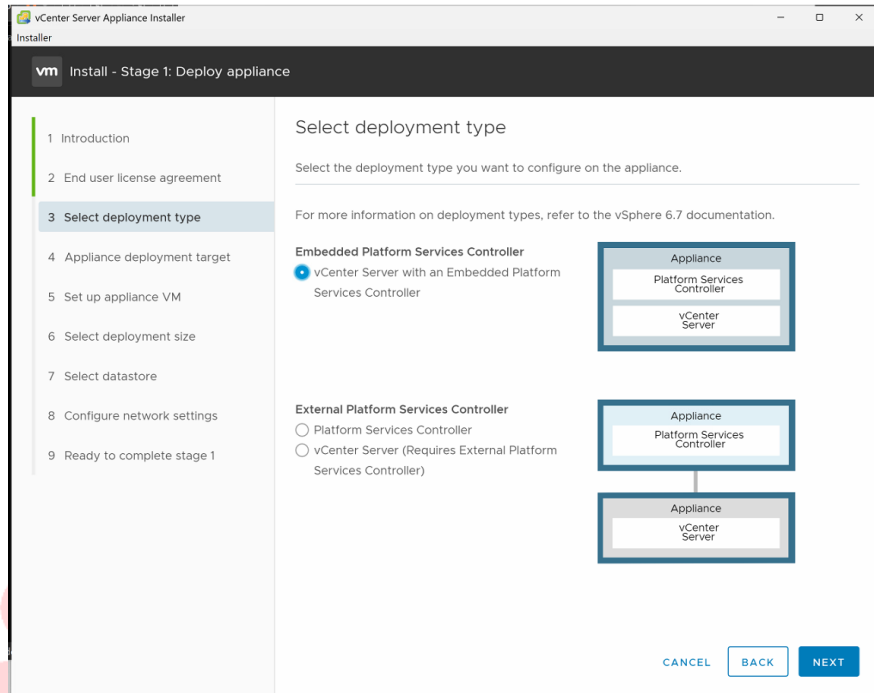
1- Selected Install



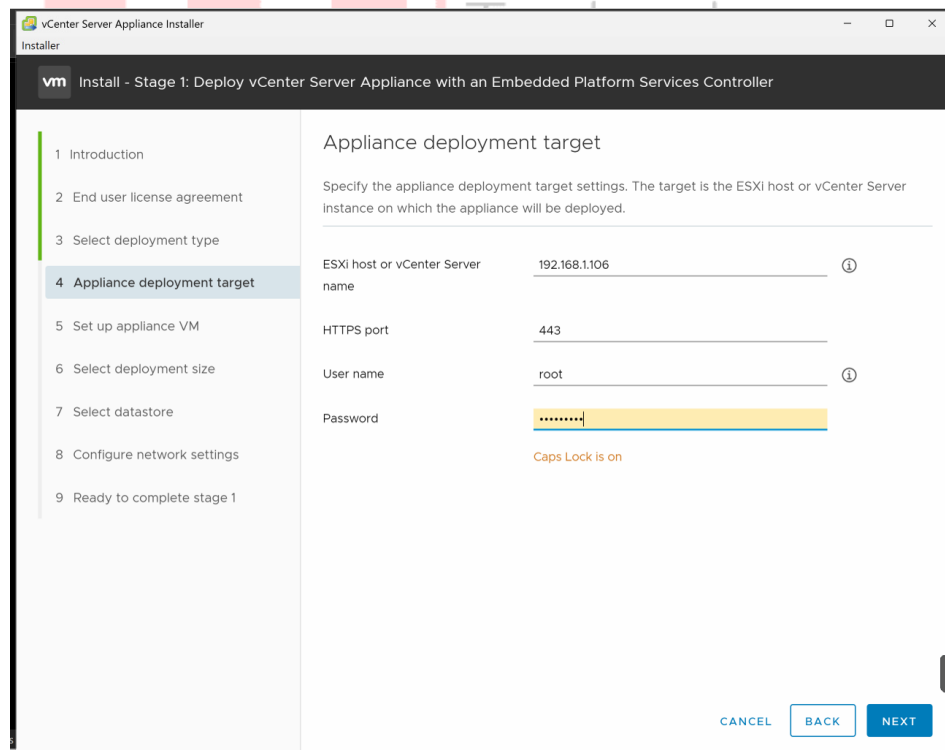
2- Accepted the terms of the license agreement



3- Selected vCenter with Embedded PSC.



4- Selected the Appliance deployment target or the ESXi host in which the vCenter will be deployed on which had an IP address of 192.168.1.106.



5- Setting up the appliance VM including the name of the vCenter VM and setting the root password as Linux\$123

The screenshot shows the 'vCenter Server Appliance Installer' window at 'Stage 1: Deploy vCenter Server Appliance with an Embedded Platform Services Controller'. The left sidebar lists steps 1 through 9, with '5 Set up appliance VM' selected. The main area is titled 'Set up appliance VM' and contains the instruction 'Specify the VM settings for the appliance to be deployed.' Below this are three input fields: 'VM name' with the value 'VMware vCenter Server Appliance', 'Set root password' (empty), and 'Confirm root password' (empty). Both password fields have a 'Caps Lock is on' warning. At the bottom right are 'CANCEL', 'BACK', and 'NEXT' buttons.

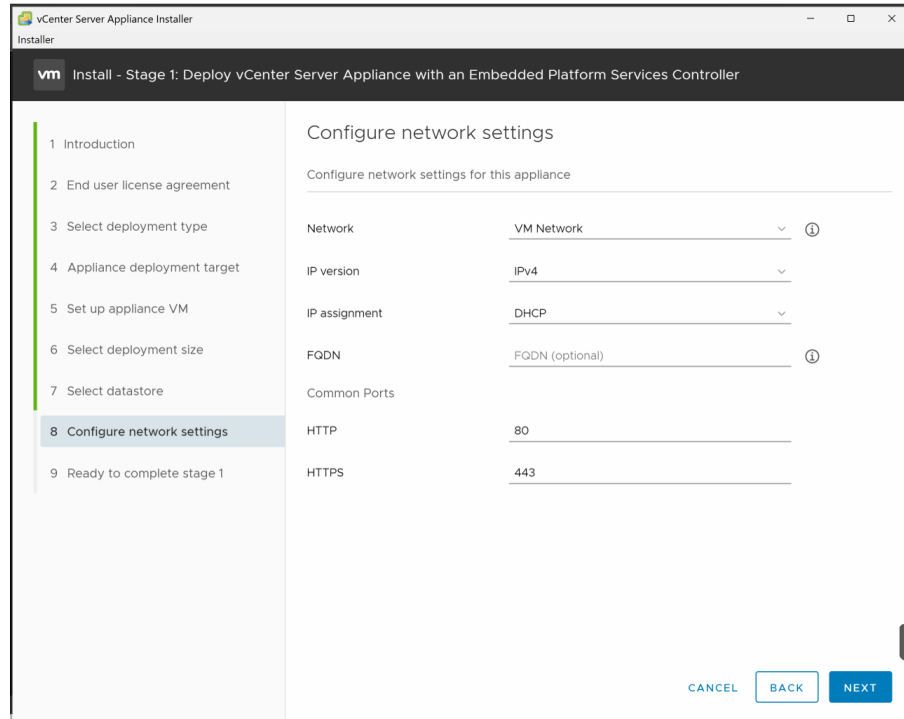
6- Selecting the datastore and enabling thin disk mode so that storage space is allocated on demand, not upfront

The screenshot shows the 'vCenter Server Appliance Installer' window at 'Stage 1: Deploy vCenter Server Appliance with an Embedded Platform Services Controller'. The left sidebar lists steps 1 through 9, with '7 Select datastore' selected. The main area is titled 'Select datastore' and contains the instruction 'Select the storage location for this appliance'. There are two radio button options: 'Install on an existing datastore accessible from the target host' (selected) and 'Install on a new vSAN cluster containing the target host'. Below the first option is a table with the following data:

Name	Type	Capacity	Free	Provisioned	Thin Provisioning
datastore1	VMFS-6	52.5 GB	51.09 GB	1.41 GB	Supported

Below the table, it says '1 item'. There is a checkbox 'Enable Thin Disk Mode' which is checked. At the bottom right are 'CANCEL', 'BACK', and 'NEXT' buttons.

7-Configuring the network setting by selecting dynamic IP assignment

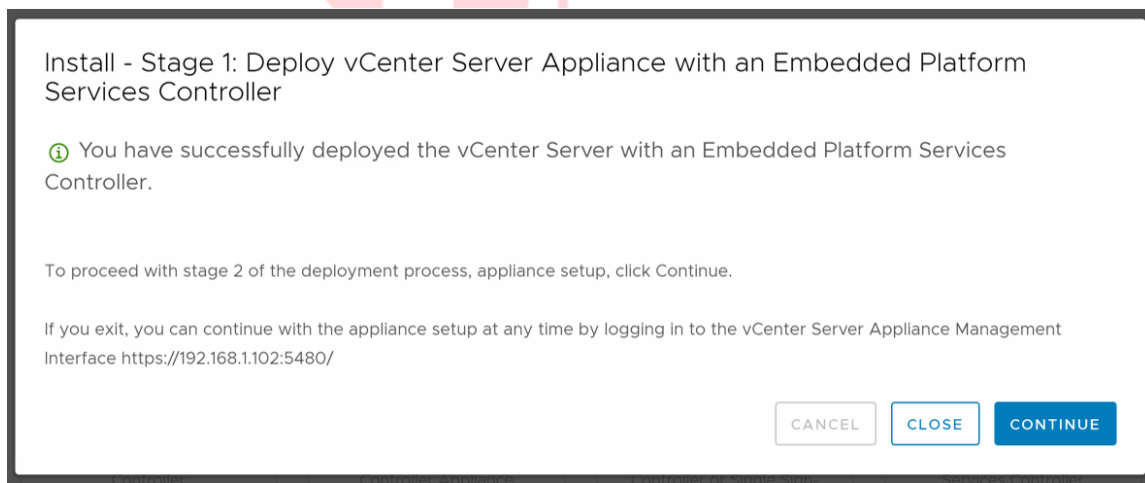


The screenshot shows the 'vCenter Server Appliance Installer' window at 'Install - Stage 1: Deploy vCenter Server Appliance with an Embedded Platform Services Controller'. The left sidebar lists steps 1 through 9, with step 8 'Configure network settings' highlighted. The main area is titled 'Configure network settings' and contains the following configuration options:

Setting	Value
Network	VM Network
IP version	IPv4
IP assignment	DHCP
FQDN	FQDN (optional)
Common Ports	
HTTP	80
HTTPS	443

At the bottom right, there are three buttons: 'CANCEL', 'BACK', and 'NEXT'.

8- Summarizing all of the configuration and continuing to Stage 1



The screenshot shows the summary screen for 'Install - Stage 1: Deploy vCenter Server Appliance with an Embedded Platform Services Controller'. It contains the following text:

Install - Stage 1: Deploy vCenter Server Appliance with an Embedded Platform Services Controller

① You have successfully deployed the vCenter Server with an Embedded Platform Services Controller.

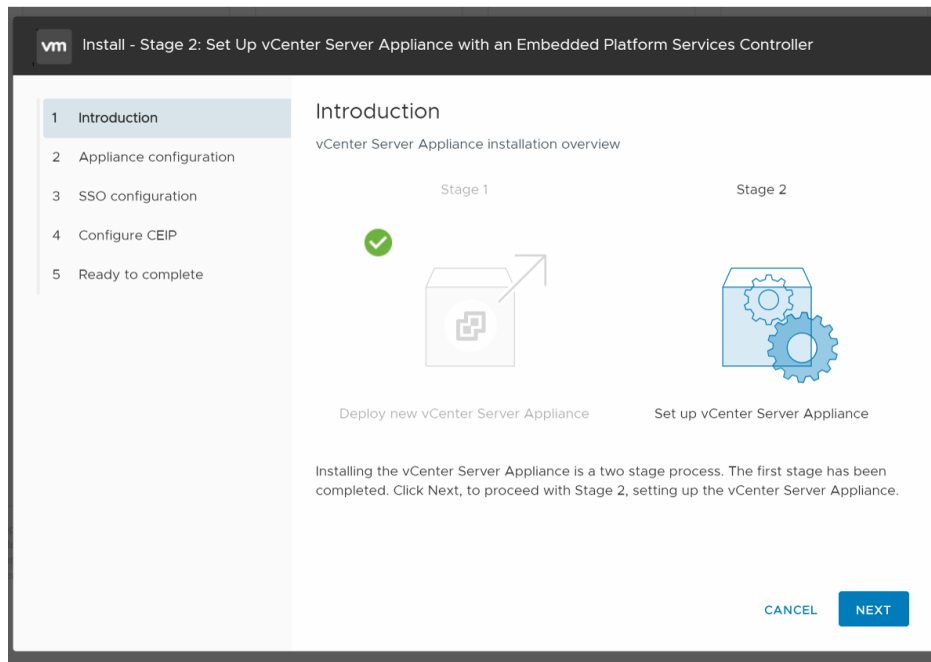
To proceed with stage 2 of the deployment process, appliance setup, click Continue.

If you exit, you can continue with the appliance setup at any time by logging in to the vCenter Server Appliance Management Interface <https://192.168.1.102:5480/>

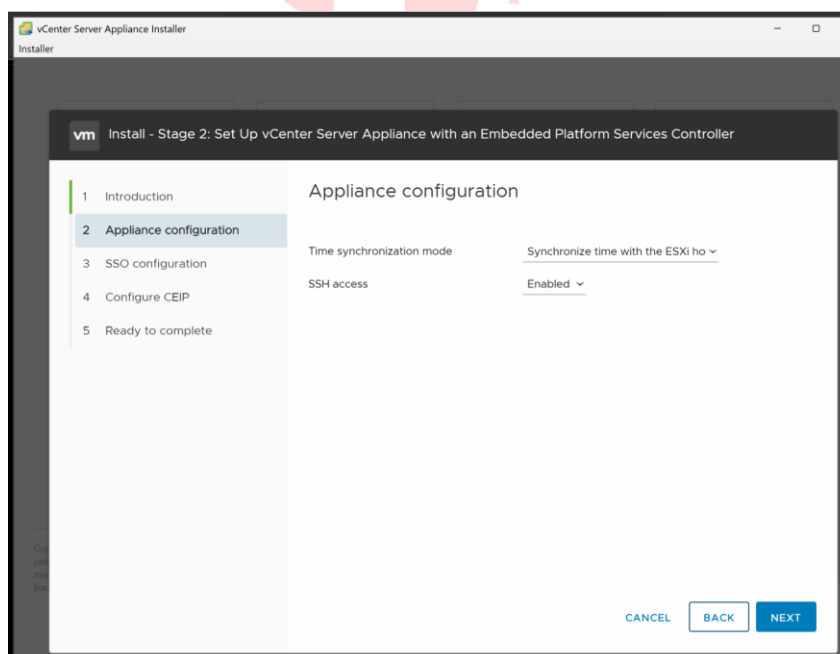
At the bottom right, there are three buttons: 'CANCEL', 'CLOSE', and 'CONTINUE'.

Stage-2:

1- Setting up the vCenter Server Appliance



2- Configuring the Appliance by selecting the time synchronization mode of the vCenter server with ESXi host and enabling SSH access.



3- Creating a new SSO domain in SSO configuration since this is the vCenter that is deployed in this environment

vm Install - Stage 2: Set Up vCenter Server Appliance with an Embedded Platform Services Controller

1 Introduction
2 Appliance configuration
3 SSO configuration
4 Configure CEIP
5 Ready to complete

SSO configuration

☒ Create a new SSO domain

Single Sign-On domain name vsphere.local ⓘ

Single Sign-On user name administrator

Single Sign-On password ⓘ

Confirm password
Caps Lock is on

☐ Join an existing SSO domain

CANCEL BACK NEXT

4- Configuring CEIP by accepting

vm Install - Stage 2: Set Up vCenter Server Appliance with an Embedded Platform Services Controller

1 Introduction
2 Appliance configuration
3 SSO configuration
4 Configure CEIP
5 Ready to complete

Configure CEIP

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Additional information regarding the data collected through CEIP and the purposes for which it is used by VMware is set forth in the Trust & Assurance Center at <http://www.vmware.com/trustvmware/ceip.html>.

If you prefer not to participate in VMware's CEIP for this product, you should uncheck the box below. You may join or leave VMware's CEIP for this product at any time.

☒ Join the VMware's Customer Experience Improvement Program (CEIP)

CANCEL BACK NEXT

5- Summarizing the vCenter Server setup

vm Install - Stage 2: Set Up vCenter Server Appliance with an Embedded Platform Services Controller

1 Introduction

2 Appliance configuration

3 SSO configuration

4 Configure CEIP

5 Ready to complete

Ready to complete

Review your settings before finishing the wizard.

Network Details	
Network configuration	Continue with the existing settings
Appliance Details	
Time synchronization mode	Synchronize time with the ESXi host
SSH access	Enabled
SSO Details	
Domain name	vsphere.local
User name	administrator
Customer Experience Improvement Program	
CEIP setting	Opted in

CANCEL BACK FINISH

6- Starting the Appliance Setup

Install - Stage 2: Appliance setup is in progress



Setup is in progress. Services on this appliance are starting.

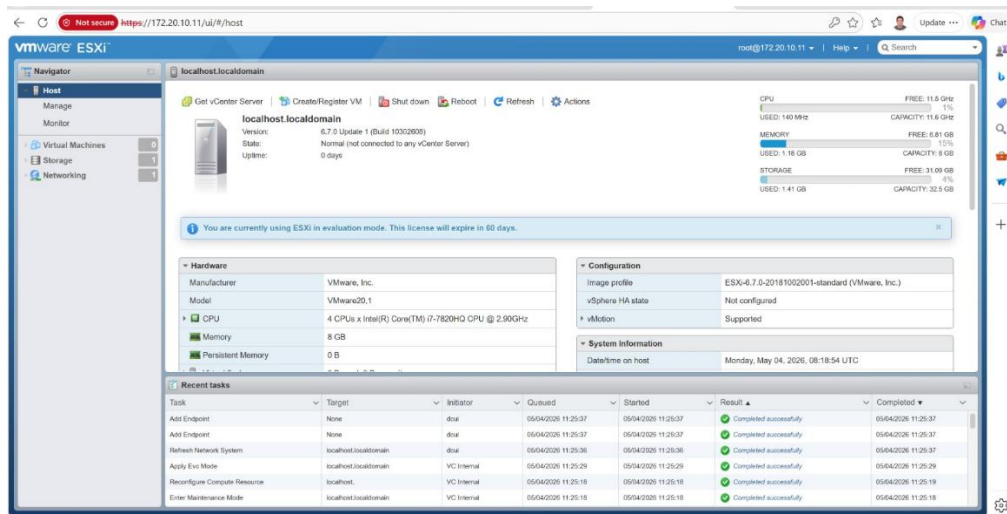
2%

Starting VMware Authentication Framework...

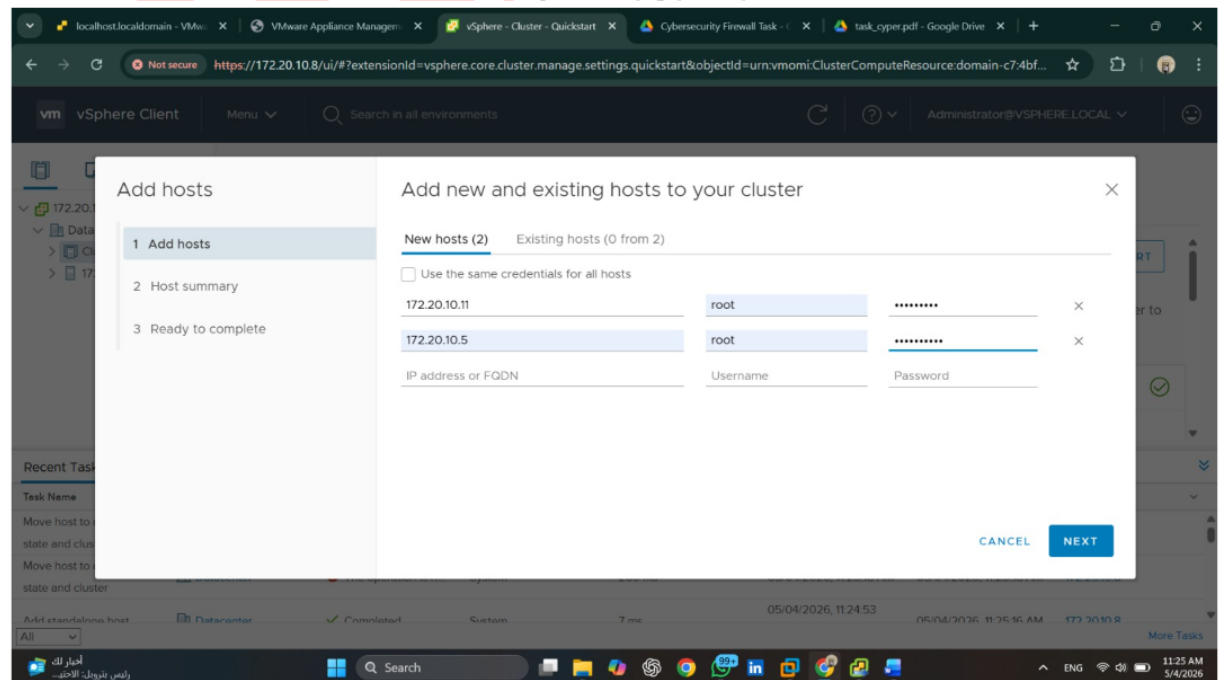
3. Connect your ESXi hosts to vCenter Server

The following ESXi host with IP address 172.20.10.11. The following screenshot is taken before this ESXi host join the vCenter.

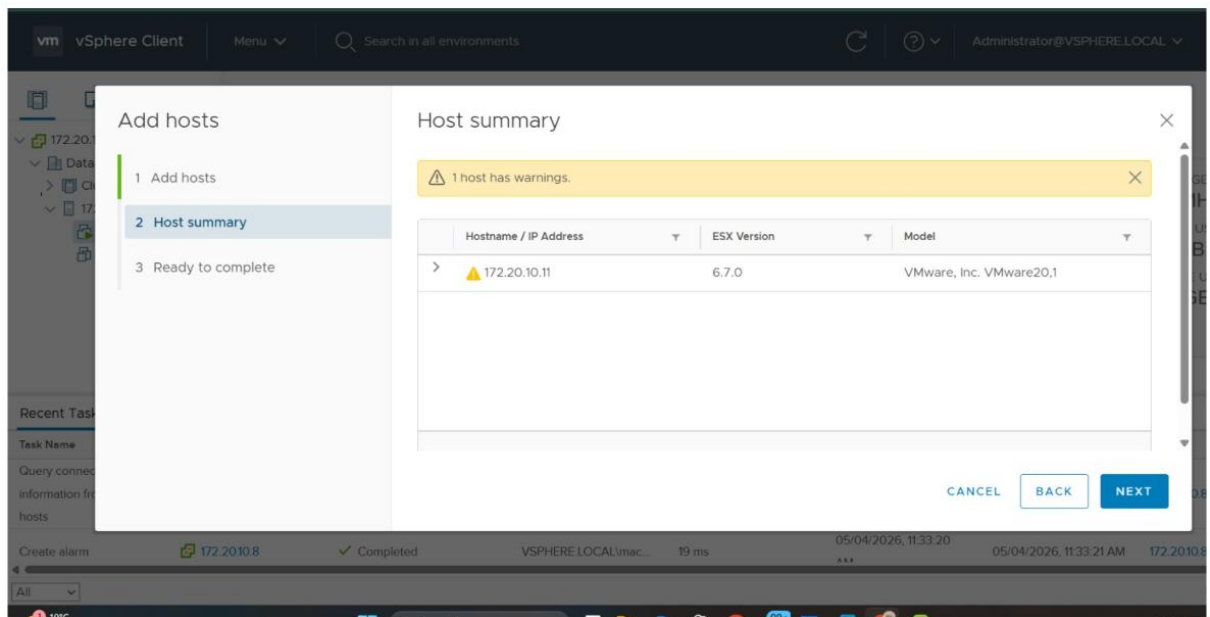
1- Before Joining the vCenter



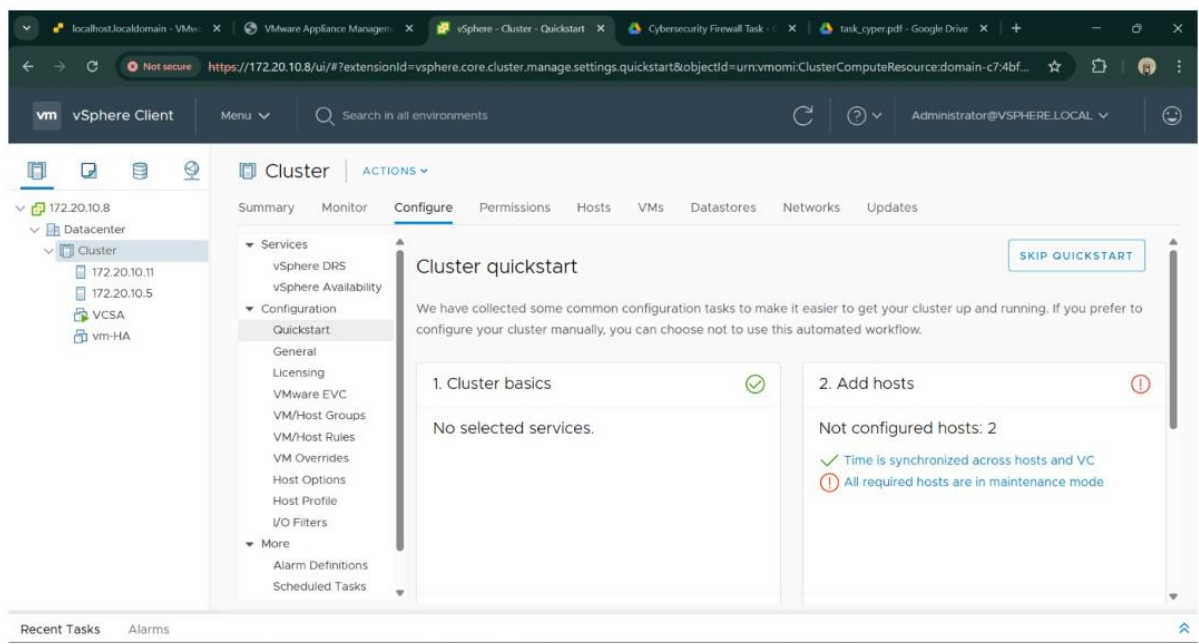
2- Joining the following ESxi hosts 172.20.10.11 and 172.20.10.5 to the vCenter



3- Host summary

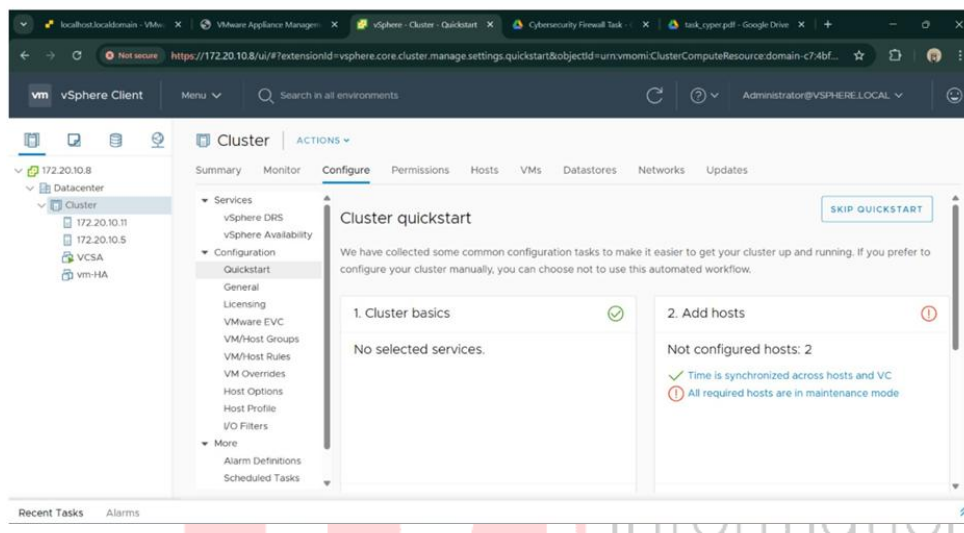


4- Joining the two ESXi hosts to the vCenter where we have also created a cluster that consist of both the ESXi hosts.

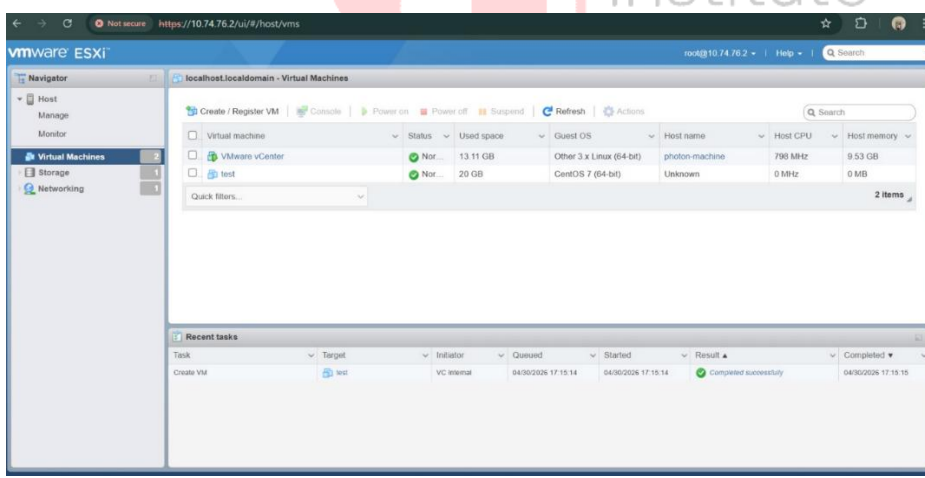


4. Combine your ESXi host into a cluster to enable advanced features like vMotion, High Availability (HA), and Distributed Resource Scheduler (DRS)

To enable vMotion, HA and DRS, we need to create a cluster consisting of the ESXi hosts. The cluster as shown consisted of two ESXi hosts 172.20.10.11 and 172.20.10.5 as shown below



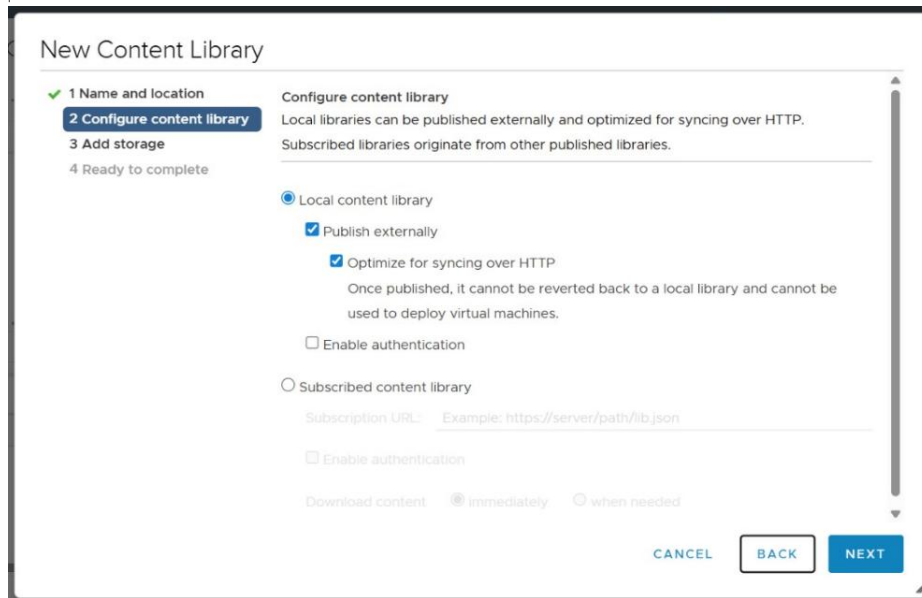
5. Create a basic virtual machine and install an operating system (Win or Linux)



Here, we have created a VM test where it has guest OS CentOS 7 (64-bit) as shown in the screenshot above.

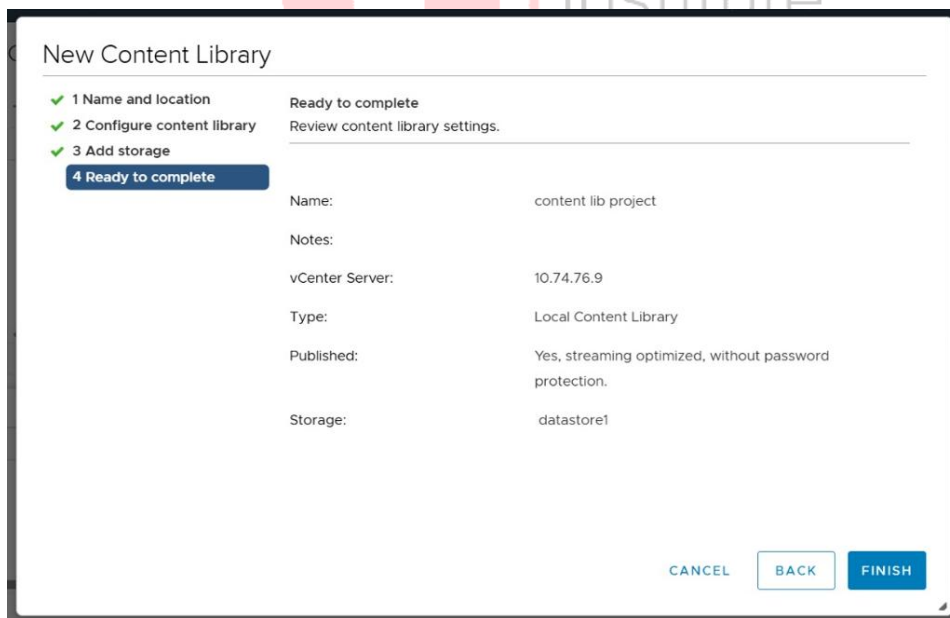
6. Upload ISO to a content library

1- We created a new Content Library as shown below where we have created a local content library as shown below



The screenshot shows the 'New Content Library' wizard at step 2, 'Configure content library'. The left sidebar shows the progress: 1 Name and location (checked), 2 Configure content library (active), 3 Add storage, and 4 Ready to complete. The main area has a title 'Configure content library' and explanatory text: 'Local libraries can be published externally and optimized for syncing over HTTP. Subscribed libraries originate from other published libraries.' There are two radio button options: 'Local content library' (selected) and 'Subscribed content library'. Under 'Local content library', there are checkboxes for 'Publish externally' (checked), 'Optimize for syncing over HTTP' (checked), and 'Enable authentication' (unchecked). A note states: 'Once published, it cannot be reverted back to a local library and cannot be used to deploy virtual machines.' Under 'Subscribed content library', there is a 'Subscription URL' field with an example 'https://server/path/lib.json', an 'Enable authentication' checkbox (unchecked), and a 'Download content' section with radio buttons for 'immediately' (selected) and 'when needed' (unchecked). At the bottom right are 'CANCEL', 'BACK', and 'NEXT' buttons.

2- The Content library storage was configured to be the local datastore 'datastore1'

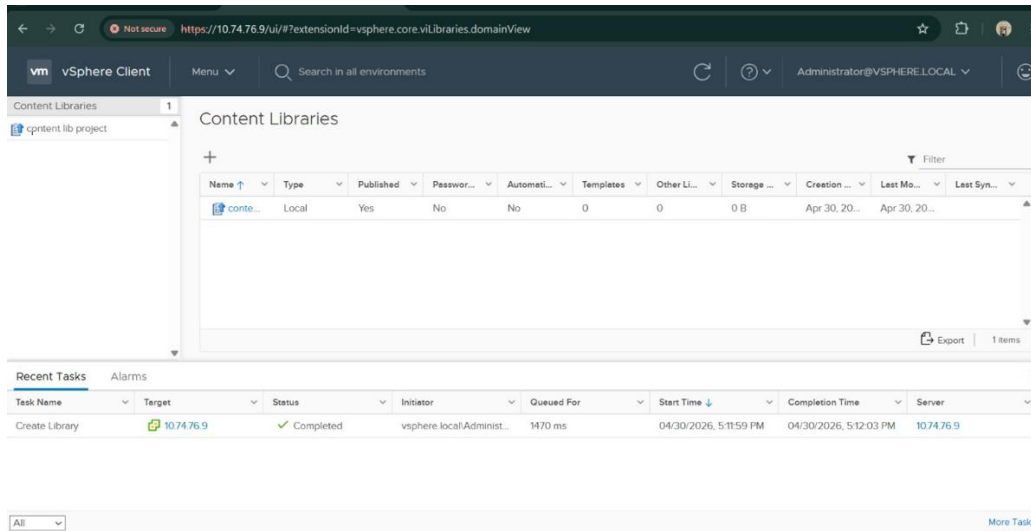


The screenshot shows the 'New Content Library' wizard at step 4, 'Ready to complete'. The left sidebar shows the progress: 1 Name and location (checked), 2 Configure content library (checked), 3 Add storage (checked), and 4 Ready to complete (active). The main area has a title 'New Content Library' and a subtitle 'Ready to complete'. Below the subtitle is the text 'Review content library settings.' A table displays the final configuration details:

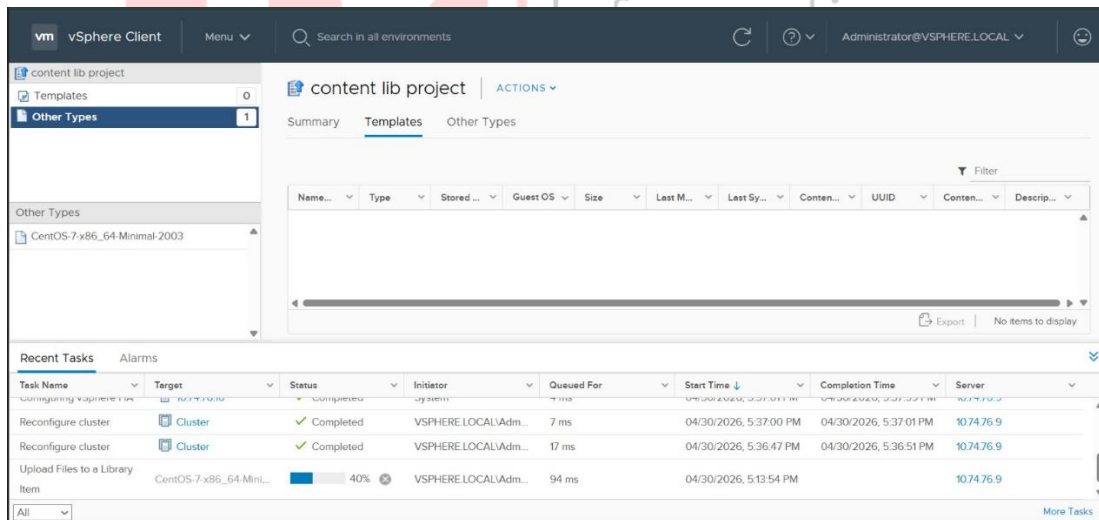
Name:	content lib project
Notes:	
vCenter Server:	10.74.76.9
Type:	Local Content Library
Published:	Yes, streaming optimized, without password protection.
Storage:	datastore1

At the bottom right are 'CANCEL', 'BACK', and 'FINISH' buttons.

3- Here the content library was created



4- Here we have uploaded the ISO to the Content Library



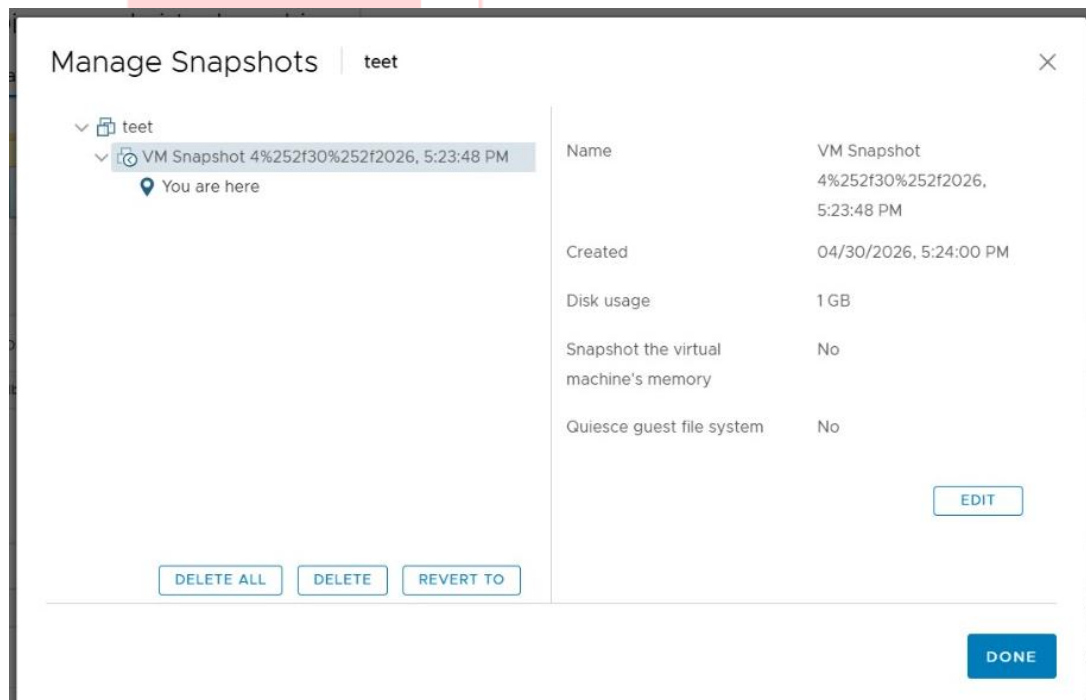
7. Perform operations like cloning, snapshotting, and template creation on your VM

➤ For snapshot

For the snapshot, we have done the following

- Right-clicked on the VM in the inventory
- Select snapshot
- Choose 'Take Snapshot'

The following is the snapshot taken



➤ For Cloning

Here we have followed the steps of cloning existing Virtual Machine. The Virtual Machine is 'teet'

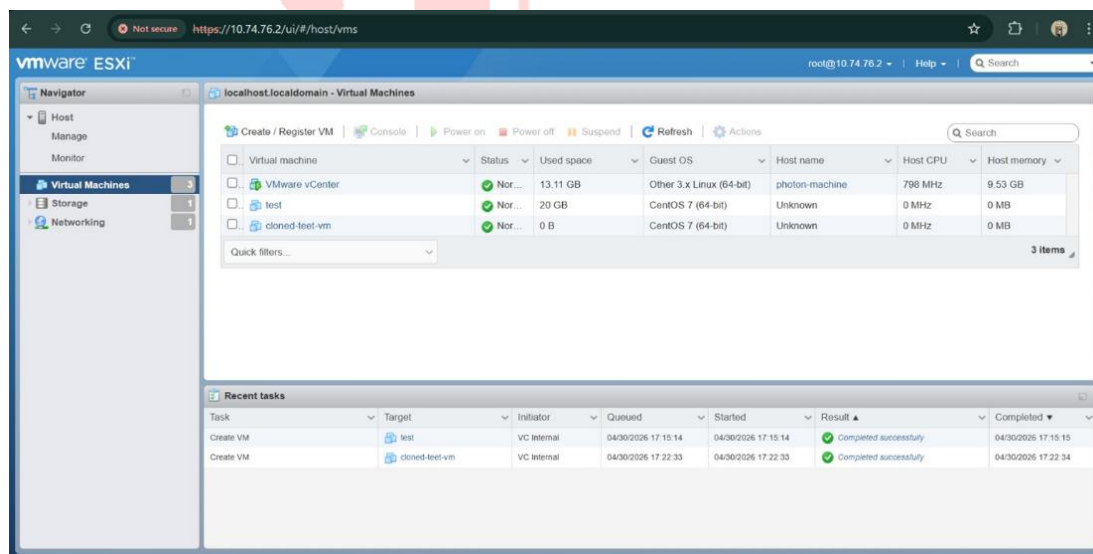
teet - Clone Existing Virtual Machine

✓ 1 Select a name and folder
✓ 2 Select a compute resource
✓ 3 Select storage
✓ 4 Select clone options
5 Ready to complete

Ready to complete
Click Finish to start creation.

Provisioning type	Clone an existing virtual machine
Source virtual machine	teet
Virtual machine name	cloned-teet-vm
Folder	project
Host	10.74.76.2
Datastore	datastore1
Disk storage	Same format as source

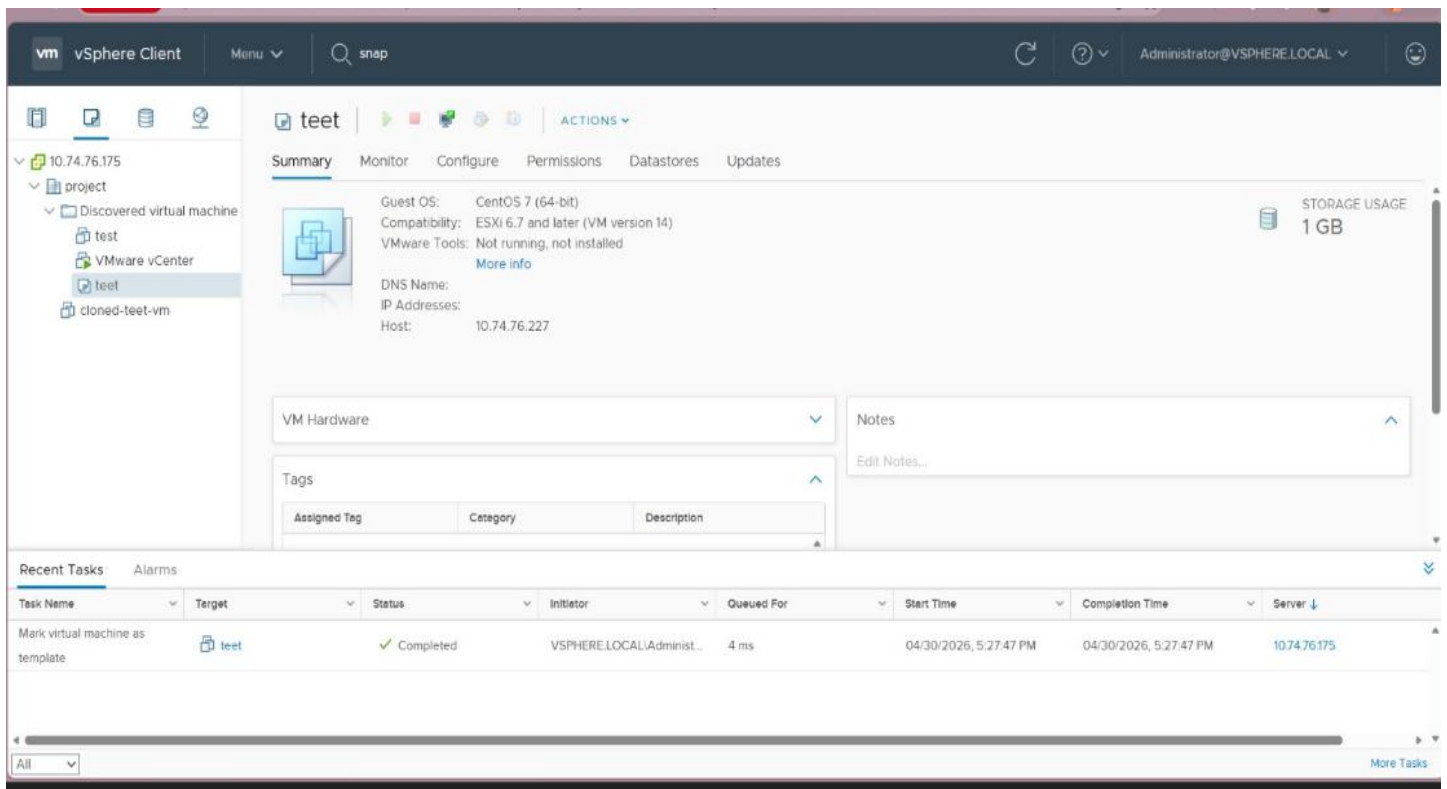
CANCEL BACK FINISH



➤ For Template

We have converted VM to template using the following steps

- Power off the teet VM
- Right click on the VM, selected Template then selected convert to Template

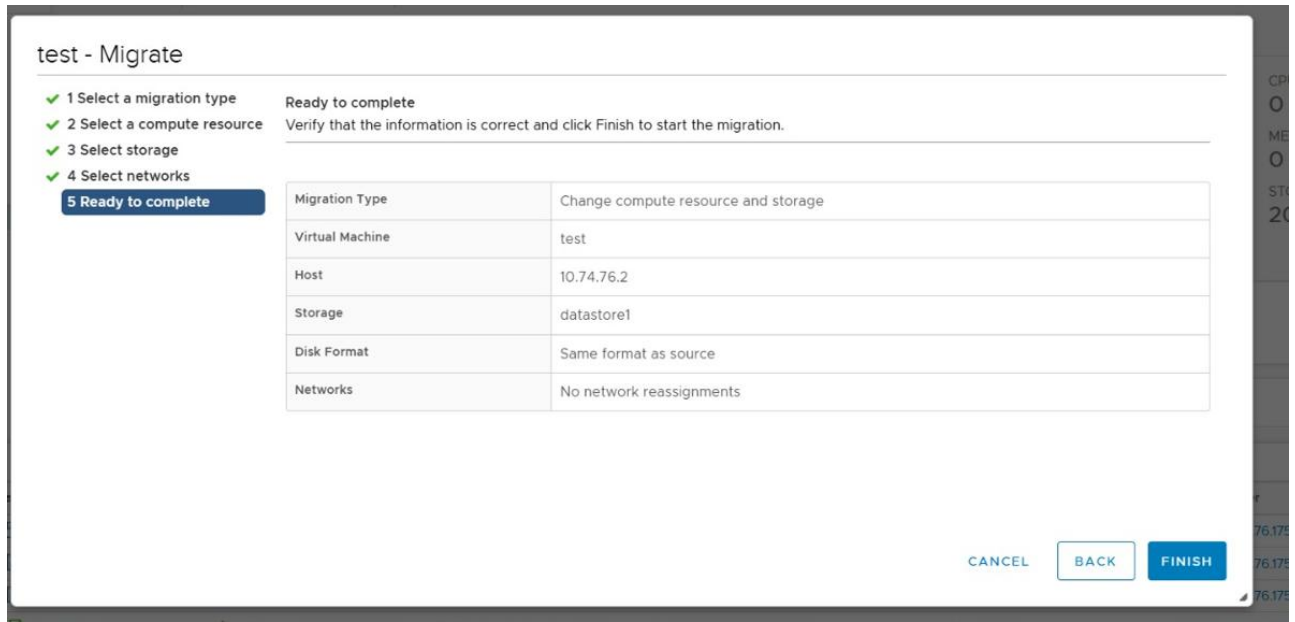


As shown under recent tasks that virtual machine teet was marked as a template.

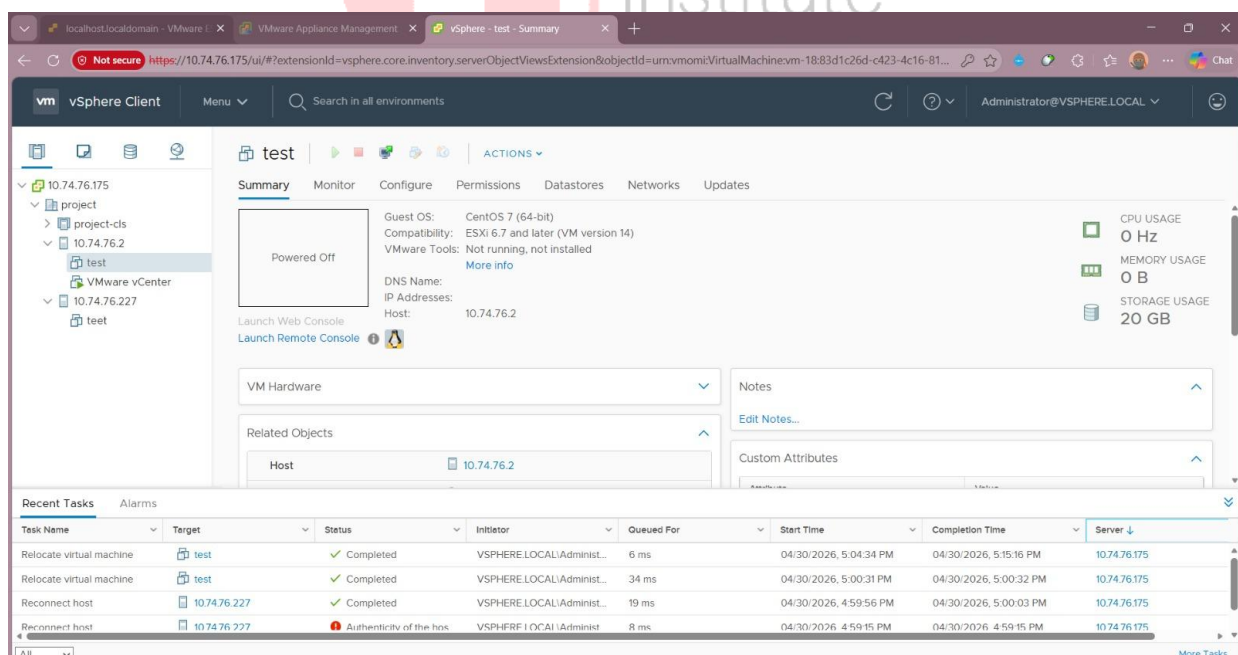
8. Migrating the VMs between ESXi hosts

Here we have made sure that the shared storage is seen between the Source and the destination host.

We proceeded with migrating test VM

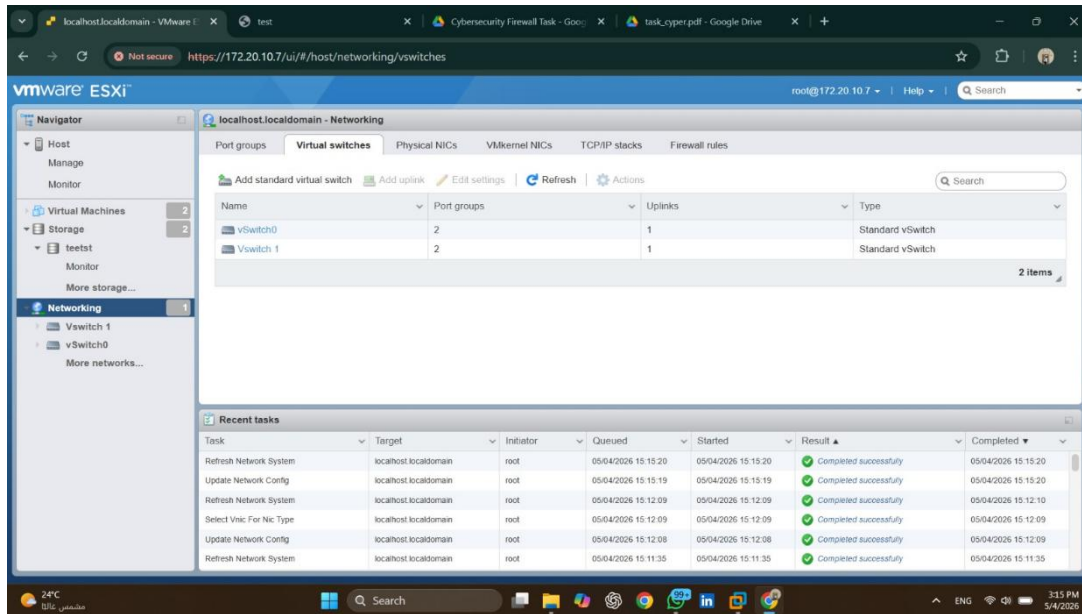


Here it is shown that test VM was relocated from one host to another under recent tasks



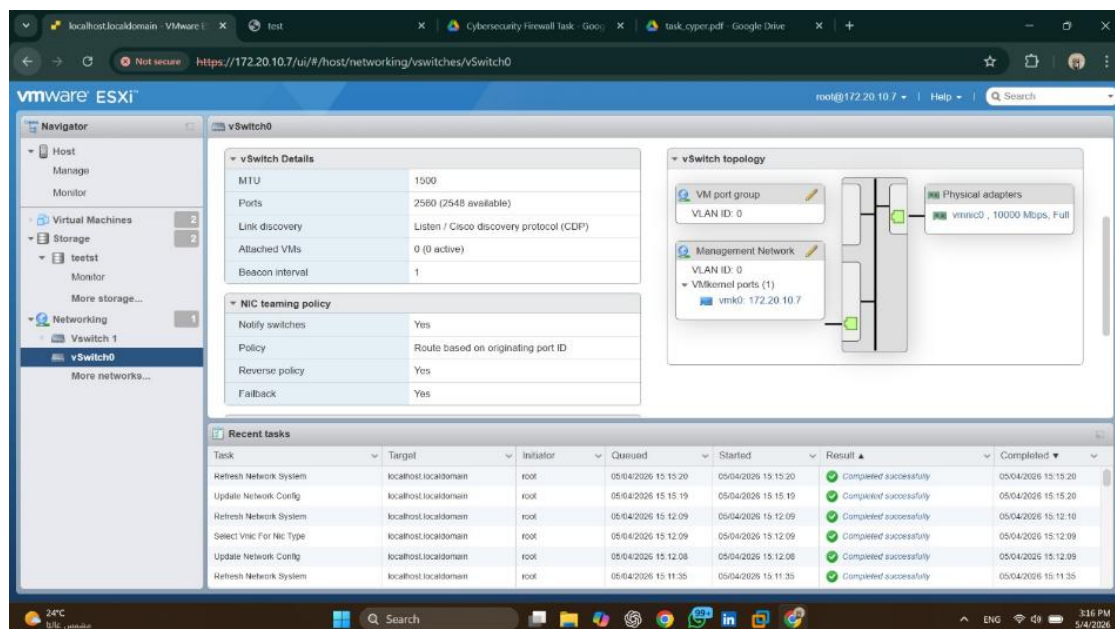
9. Create and configure virtual switches (vSwitches), one for management and VMs port groups, and another switch for Storage and Migration

Here we have created two vSwitches. vSwitch 0 and vSwitch 1

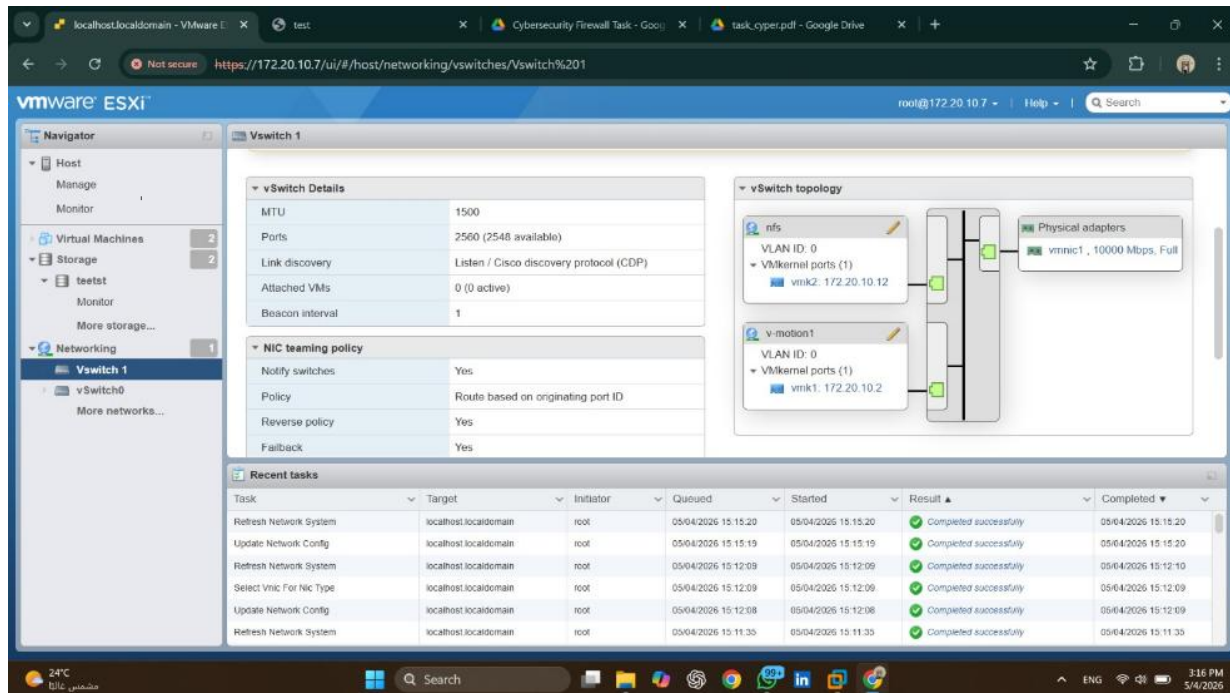


Here as shown under recent tasks that test VM has been migrated from one host to another host.

For vSwitch 0, it is configured for Management network and VM port group.

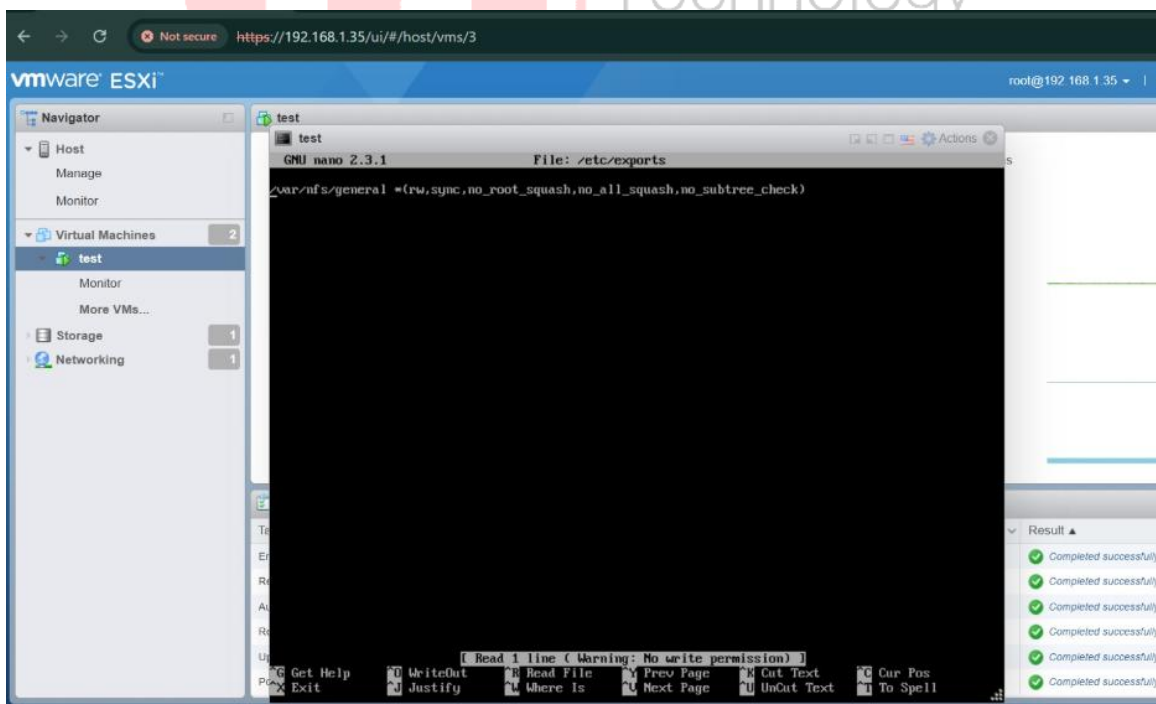


For vSwitch1, it is configured for Storage and Migration



10. Create and manage NFS datastore

Here, we have edited the /etc/exports file as follows



11. Configure HA and DRS, try to make one esxi fails, to make sure if the VM will start on the other esxi

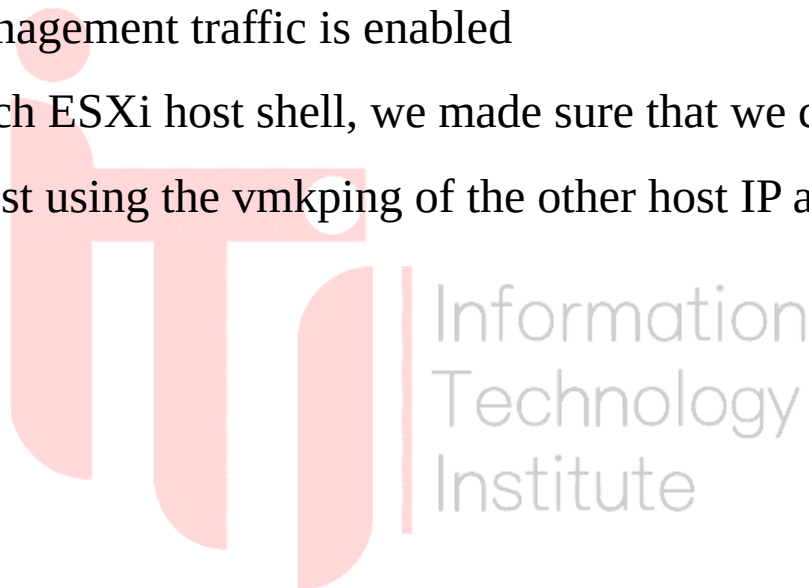
To enable both HA and DRS, we selected vSphere DRS and vSphere Availability and turned both the DRS and HA ON.

➤ **For High Availability (HA):** we did the following prerequisites.

1- We configured the Management Network where each host had it's own

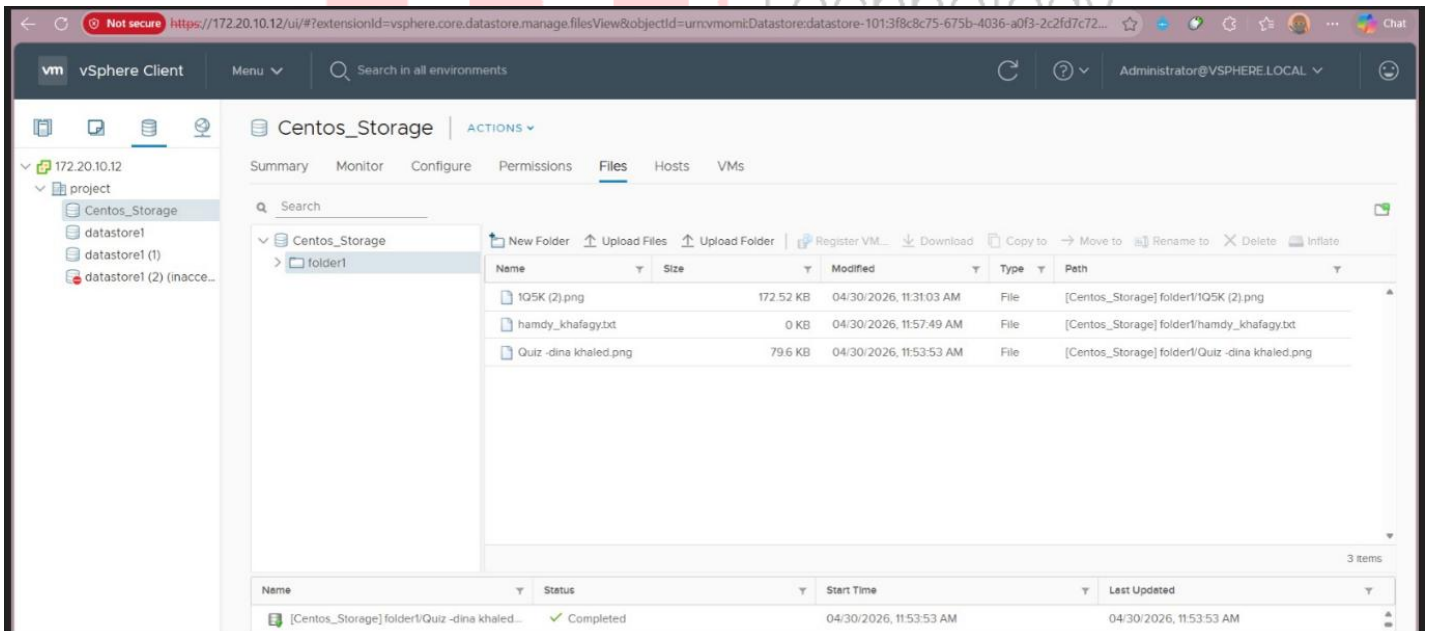
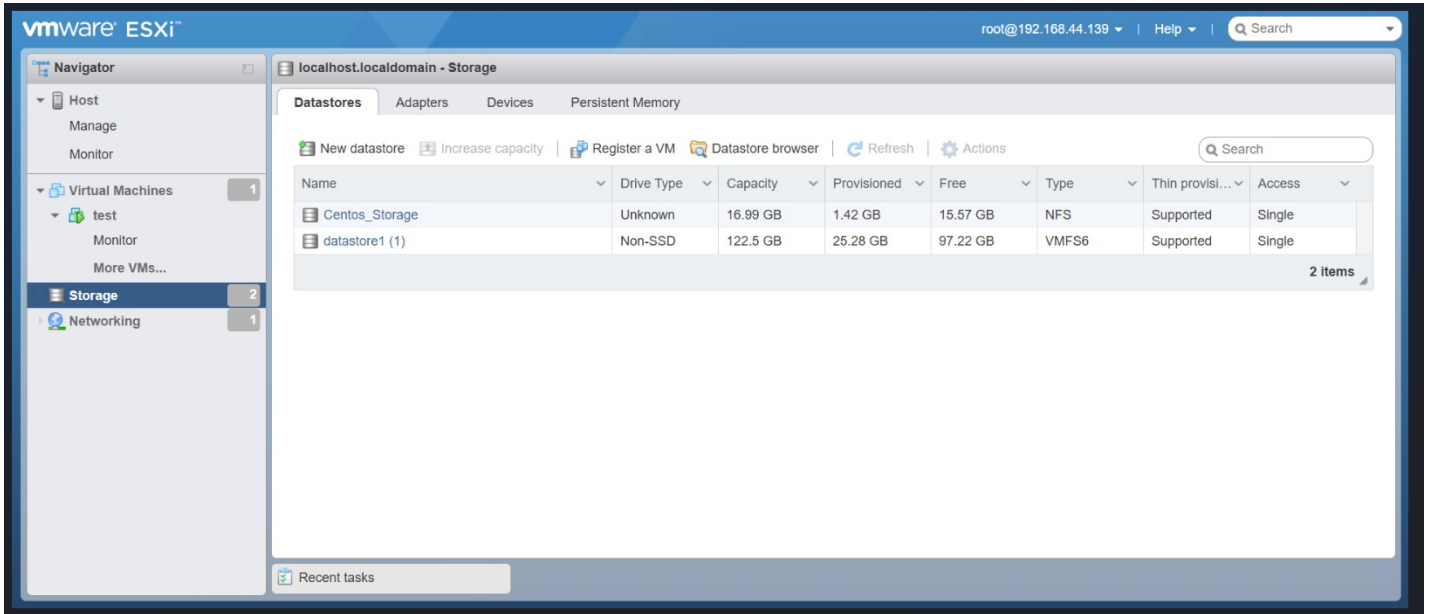
- Vmkernel adapter (vmk0)
- Management traffic is enabled

From each ESXi host shell, we made sure that we can ping on the Other host using the vmkping of the other host IP address.



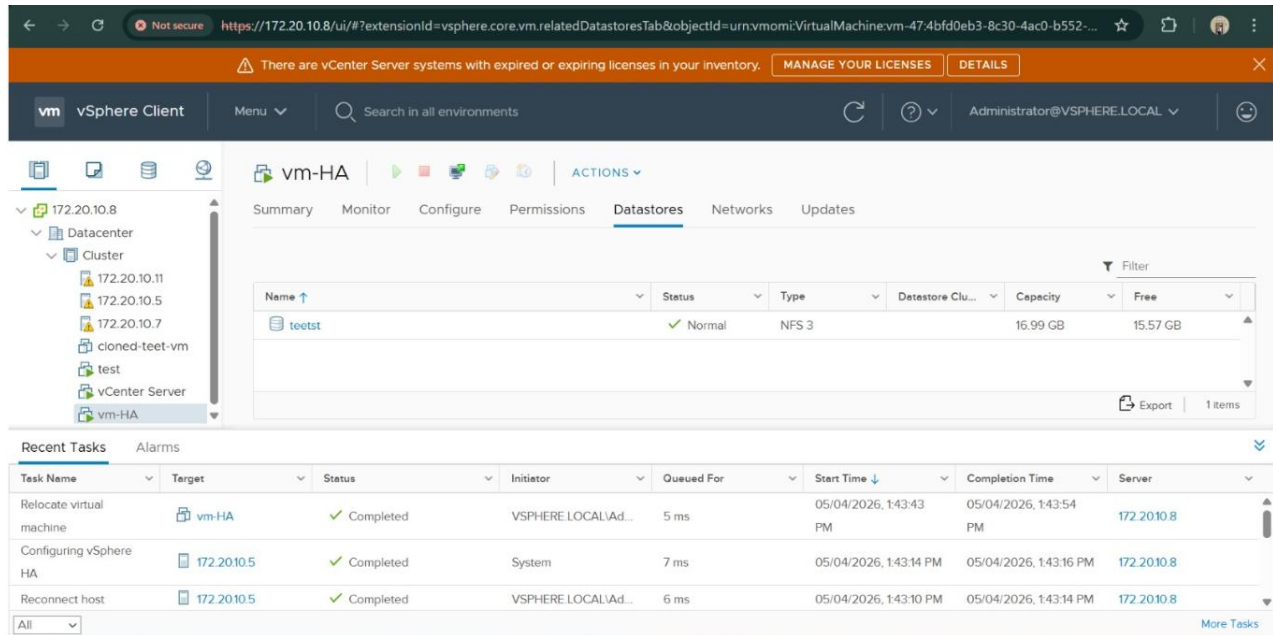
2- We have created a shared storage between the two ESXi hosts called CentOS storage where

- The newly created VM files are stored on the shared storage.
- The shared datastore is mounted on both ESXi hosts in the cluster.



3- We enabled HA on the Cluster

4- We powered off an ESXi host to simulate an ESXi host failure so that the VM will be restarted on the other ESXi host as shown below.



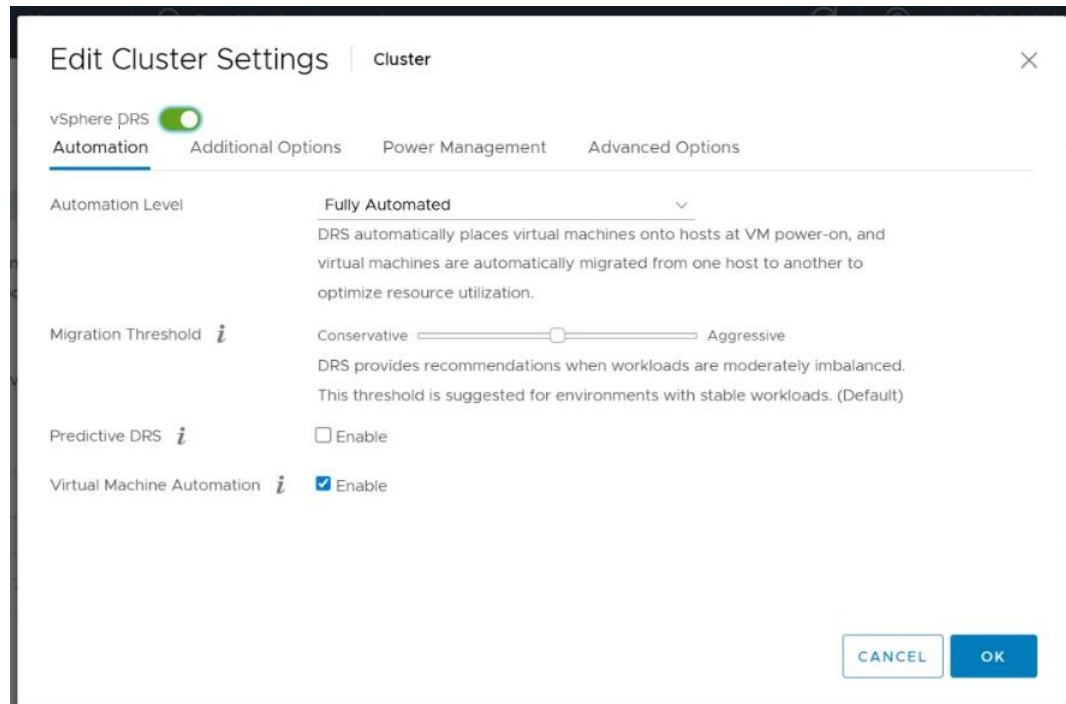
The screenshot displays the vSphere Client interface. The left sidebar shows the navigation tree with 'vm-HA' selected under the 'vCenter Server' node. The main pane shows the 'Datastores' tab for 'vm-HA'. A table lists the datastore 'teetst' with a status of 'Normal', type 'NFS 3', capacity '16.99 GB', and free space '15.57 GB'. Below this, the 'Recent Tasks' section shows a list of tasks:

Task Name	Target	Status	Initiator	Queued For	Start Time	Completion Time	Server
Relocate virtual machine	vm-HA	Completed	VSPHERE LOCAL/Ad...	5 ms	05/04/2026, 1:43:43 PM	05/04/2026, 1:43:54 PM	172.20.10.8
Configuring vSphere HA	172.20.10.5	Completed	System	7 ms	05/04/2026, 1:43:14 PM	05/04/2026, 1:43:16 PM	172.20.10.8
Reconnect host	172.20.10.5	Completed	VSPHERE LOCAL/Ad...	6 ms	05/04/2026, 1:43:10 PM	05/04/2026, 1:43:14 PM	172.20.10.8

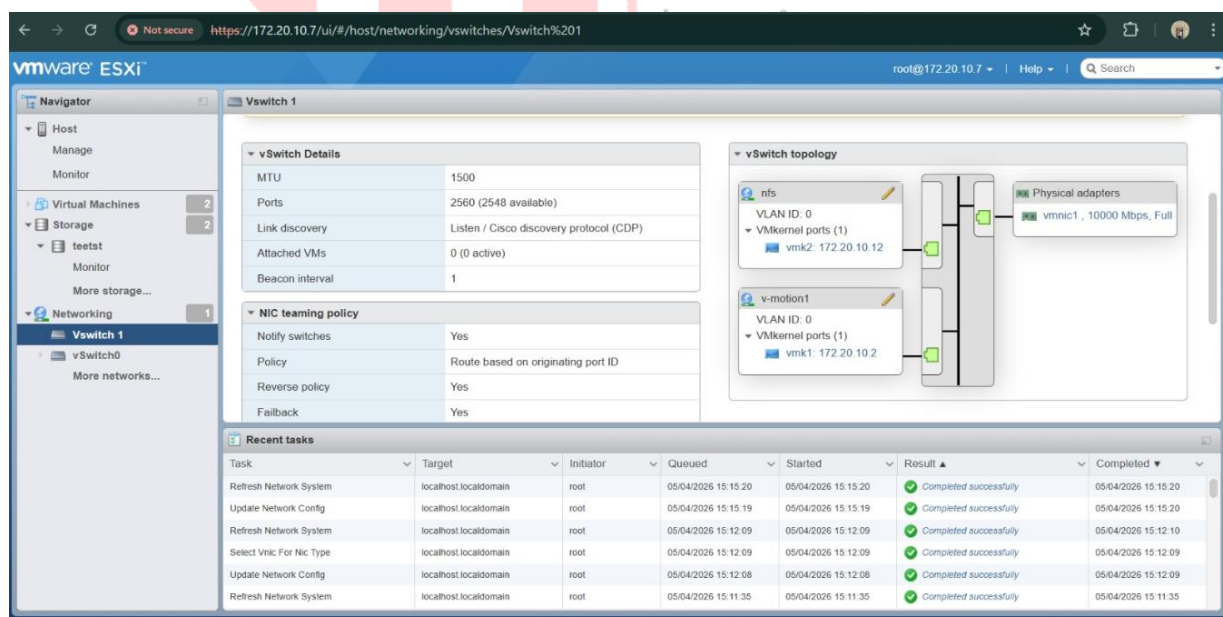
Note: Due to an issue that occurred in the shared storage, we have recreated a new one with name teetst and made sure that the VM-HA which was the VM that has restarted on another host can access the Shared datastore

➤ For Distributed Resource Scheduler (DRS):

1- We have enabled the DRS as shown below



2- We made sure that there are two Vmkernel adapters enabled, one for the management vmk0 which was already enabled and vmk1 for vmotion.

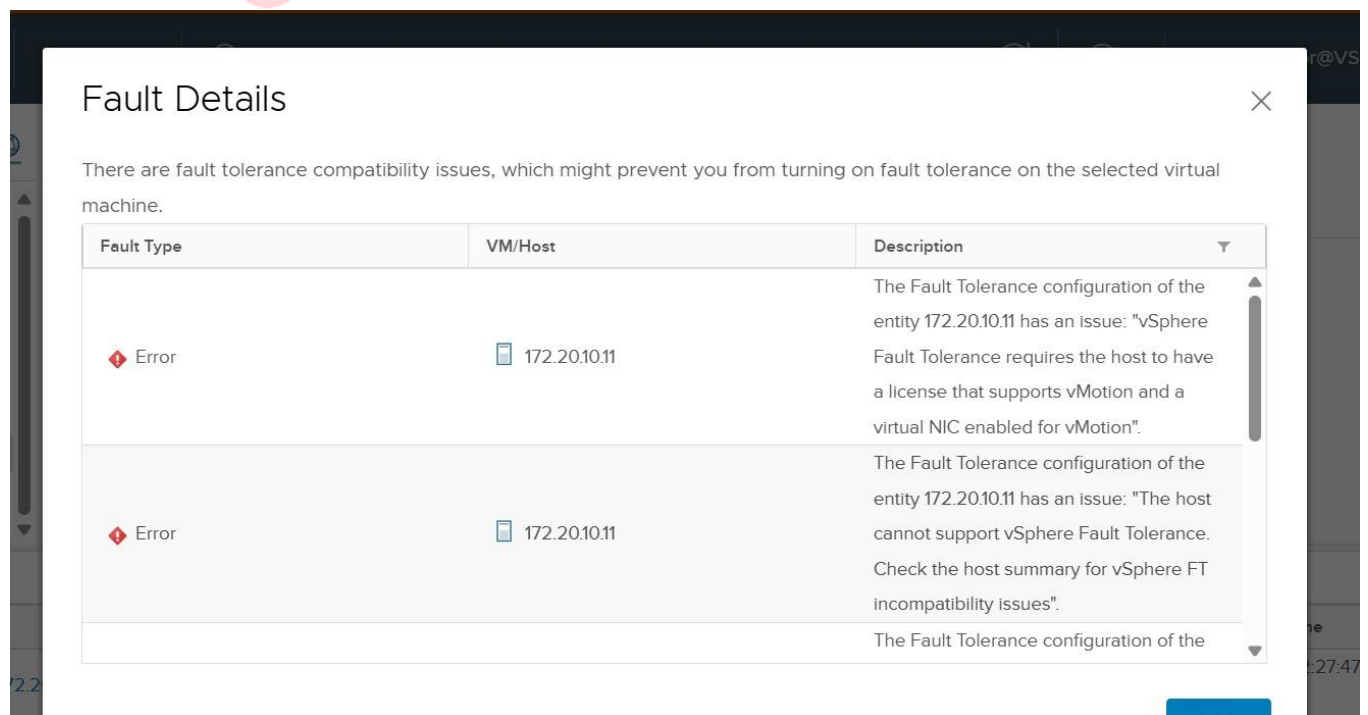


Here as displayed in the vSwitch topology, VMK1 is created for one of the esxi hosts in the cluster.

In this case, The DRS will automatically place virtual machine to the under utilized esxi host so that resources consumption are balanced between both.

12. Enable FT for your VM

For the fault tolerance which provides zero downtime and zero data loss by running a secondary VM in lockstep with the primary VM . The Fault tolerance will depend on FT logging network and the Shared datastore to be implemented



Here, we have encountered an Issue due to license that is absent as the following screenshot shows